COMP3050 Individual Dissertation Single Honours

Prompt-engineering-based Hallucination Estimation for large language models

Yuyang ZHANG

20514470

Supervised by Qiao Lin

Programme of Study: CS

April 13, 2026

School of Computer Science

University of Nottingham Ningbo China

Abstract

Large Language Models (LLMs) have achieved remarkable performance across a wide range of natural language processing tasks. However, LLMs often generate factually incorrect or fabricated information, a phenomenon commonly referred to as hallucination. While existing research has primarily focused on hallucination detection, effective mitigation strategies—particularly in open-domain question answering settings where explicit grounding is limited—remain challenging.

This project presents a multi-signal framework for hallucination detection and estimation, coupled with detector-guided mitigation strategies, including re-prompting, candidate filtering, and abstention. The proposed system integrates multiple complementary signals, including LLM-as-a-judge evaluation, natural language inference (NLI) for factual consistency, and SelfCheck-based response consistency, to produce a unified hallucination score. Based on this score, a detector-conditioned re-prompting strategy is applied to selectively revise responses or abstain when confidence is low. The framework is evaluated on multiple benchmark datasets, including TruthfulQA, FEVER, and Natural Questions, using both performance and stability metrics.

Experimental results show that the approach is most effective on evidence-grounded datasets such as FEVER, while gains on more open-ended benchmarks are limited. In particular, notable reductions in hallucination rates are observed on evidence-based datasets such as FEVER. However, performance improvements are less consistent on open-ended datasets, highlighting the challenges of mitigating hallucinations in the absence of explicit grounding.

Overall, this work highlights the importance of integrating multiple evaluation signals with adaptive mitigation strategies and shows that the effectiveness of hallucination mitigation is strongly dependent on the availability of grounding evidence.

Contents

Chapter 1

Introduction

1.1 Background

In recent years, large language models (LLMs) have demonstrated outstanding performance in various natural language processing tasks, including question answering, text summarization, language inference, knowledge retrieval, and dialogue generation [1]. These models are capable of generating fluent and context-coherent responses, thus finding widespread application in research and practical applications, such as clinical decision support systems [2]. Despite these advances, LLMs are known to generate factually incorrect or fabricated information that is not supported by the input or external knowledge, a phenomenon often referred to as "hallucination" [3]. These hallucinations can occur even when model outputs appear linguistically plausible, making them difficult to detect and potentially misleading to users. As a result, they can degrade system performance and fail to meet user expectations [4].

1.2 Motivation

Hallucinations in large language models pose a serious challenge to their reliability and practical deployment. In real-world applications, users often perceive model outputs as reliable sources of information, making hallucinated outputs highly misleading in real-world use. These errors can lead not only to distorted information dissemination, decreased user trust, and reduced system performance, but also to erroneous task execution in LLM-based agent systems, further impacting overall system reliability [5]. In high-risk applications such as medical decision support, while LLM-assisted tools can effectively support goal elicitation, clarification, and revision, the lack of human interaction, personalized patient contextual information, and the significant risk of overload due to unconstrained information access pose significant challenges [6]. These issues can amplify uncertainty and potential harm, limiting their safe application as independent asynchronous tools in real-world scenarios.

The causes of hallucinations are multifaceted, including both data-related and model-related factors. Training data may contain quality issues, including noise, bias, outdated information, subjective content, and unverified facts, leading to inaccurate or unsubstantiated model outputs [7]. Furthermore, LLMs are often poorly calibrated and prone to overconfidence, assigning excessively high confidence levels to erroneous outputs [8]. In open-domain environments, this problem is further exacerbated by a lack of reliable underlying information, allowing models to generate seemingly plausible but actually fabricated explanations [9]. These challenges collectively underscore the need for more robust methods to reliably detect, estimate and mitigate hallucinations.

Despite the growing interest in hallucination detection, existing methods remain limited in several aspects. Many hallucination evaluation methods rely on single-model or single-signal strategies, such as confidence estimation, natural language inference (NLI), or response consistency checks [4]. While these approaches provide advantages in handling lexical variations or detecting simple inconsistencies, they are prone to error propagation from underlying models, struggle to pinpoint hallucinated substrings at a fine-grained level, and often fail to fully capture the diversity (e.g., intrinsic vs. extrinsic) and complexity of hallucinations [4]. Therefore, these methods are often unstable and lack robustness across different tasks and datasets.

Furthermore, while detection methods have been extensively studied, effective mitigation strategies remain relatively scarce. Existing mitigation methods are often heuristic or poorly integrated with detection mechanisms, limiting their effectiveness in reducing hallucinatory outputs. This challenge is particularly pronounced in knowledge-intensive task environments such as open-domain question answering, because reliable contextual information is limited and pre-trained language models struggle to access and manipulate knowledge accurately, are prone to hallucinations, and fail to provide reliable sources for predictions [9]. These limitations underscore the need to develop a more comprehensive and integrated framework that combines detection, estimation, and mitigation measures to improve the factual reliability of LLM-generated outputs.

1.3 Research Gap

Despite recent progress in hallucination detection and mitigation, several key limitations remain. Firstly, many existing hallucination detection methods tend to rely on a single signal, such as model uncertainty estimation, Natural Language Entailment (NLI), or consistency among sampled responses. This dependence on a single source of information limits their robustness across different tasks, models, and datasets [10]. Secondly, research on hallucination detection metrics and mitigation strategies has largely proceeded in parallel. The decoupling of these two processes results in a lack of cohesive intervention strategies, where detection results are not effectively integrated into the generation loop to facilitate real-time mitigation [4]. Thirdly, existing hallucination evaluation metrics and mitigation methods often exhibit limited cross-dataset generalizability. While performing well on specific benchmarks, these approaches struggle to transfer effectively to other datasets or unseen domains [4].

These limitations highlight the need for more integrated and robust approaches. In particular, there is a need for multi-signal frameworks that not only improve the reliability of hallucination detection but also enable effective, detector-guided mitigation. To address this gap, this project proposes a unified framework that integrates multi-signal evaluation with detector-conditioned intervention strategies, including re-prompting, candidate filtering, and abstention mechanisms.

1.4 Aims and Objectives

This project aims to develop a unified and robust framework for hallucination detection, estimation, and mitigation in large language models (LLMs). Specifically, it addresses the limitations of existing single-signal methods and investigates how the effectiveness of hallucination mitigation varies across tasks with different levels of grounding availability. To achieve this goal, the project follows these objectives:

- To design a multi-signal hallucination detection mechanism that integrates complementary evaluation signals, including:

    1. an LLM-as-a-judge evaluation component
    2. a SelfCheckGPT-style response consistency layer based on contradiction esti-

mation across multiple generated outputs

3. an entailment-based fact verification module using natural language inference (NLI)

- To develop a quantitative hallucination estimation method that aggregates multiple evaluation signals and provides stable and reproducible measurements, while enabling the analysis of variability and consistency across different datasets through repeated sampling and statistical analysis (e.g., mean and variance).

- To implement a detector-conditioned mitigation strategy that forms a closed-loop pipeline, incorporating techniques such as re-prompting, candidate filtering, and abstention, with response selection guided by hallucination risk scores.

- To evaluate the proposed framework across multiple benchmark datasets, including TruthfulQA, FEVER, and Natural Questions, using performance and stability metrics, and to analyse dataset-dependent performance and the role of grounding.

- To conduct ablation studies and threshold tuning to assess the contribution of individual components and optimise the overall performance of the proposed framework.

1.5 Contributions

This project makes the following contributions:

- It proposes a unified multi-signal framework that integrates complementary evaluation signals for hallucination detection, estimation, and mitigation, enhancing robustness and stability compared to single-signal approaches, while revealing trade-offs between accuracy and consistency.

- It introduces a quantitative and stability-aware hallucination estimation method that leverages repeated sampling and statistical analysis to produce more reliable and reproducible measurements.

- It develops a detector-conditioned mitigation strategy that forms a closed-loop pipeline, enabling dynamic response revision, filtering, and selection based on hallucination risk.

- It provides a comprehensive cross-dataset evaluation on TruthfulQA, FEVER, and Natural Questions, demonstrating the effectiveness and limitations of the proposed approach in both grounded and open-domain settings.

- It offers empirical insights into the behaviour of multi-signal hallucination detection, including component contributions, stability characteristics, and challenges in open-domain scenarios.

Chapter 2

Literature Review

This chapter provides an overview of existing research related to hallucination in large language models, with a particular focus on detection, estimation, and mitigation approaches. The aim is to analyse the strengths and limitations of current methods and to identify gaps that motivate the proposed framework. This review is structured as follows: Section 2.1 introduces the concept and types of hallucinations in large language models. Section 2.2 examines the underlying causes of hallucination, including both data-related and model-related factors. Section 2.3 examines existing hallucination detection methods, including confidence-based, entailment-based, and consistency-based approaches. Section 2.4 discusses hallucination estimation and uncertainty quantification techniques. Section 2.5 presents mitigation strategies, including prompt-based and retrieval-augmented approaches. Section 2.6 highlights reproducibility challenges in hallucination research. Finally, Section 2.7 summarises the key limitations of existing work and identifies the research gap addressed in this project.

2.1 Concept and Types of Hallucination in LLMs

Hallucinations in Large Language Models (LLMs) refer to the generation of content that is factually inaccurate or contains fabricated information. While LLMs can generate fluent and contextually coherent responses, such outputs may lack factual grounding. These errors are difficult to detect, mislead users, and potentially undermine their reliability [3].

Existing literature has proposed different classification methods to describe hallucination phenomena. One widely accepted method is to divide hallucinations into intrinsic and extrinsic types [11]. Intrinsic hallucinations occur when the generated content contradicts the provided input or context [12]. Conversely, extrinsic hallucinations occur when the model introduces external information that the input does not support, and this information may be fabricated or unverifiable [12]. This distinction highlights that hallucinations can stem from inconsistencies with the input or from a lack of reliable evidence. In addition, hallucinations can also be divided into factuality hallucinations and faithfulness hallucinations. Factuality hallucinations emphasize that the generated content is inconsistent with real-world facts, even if it doesn't contradict the input [11]. In contrast, faithfulness hallucinations refer to generated content lacking support from the given input or source context, or being inconsistent with it [11]. This perspective emphasizes that hallucinations can stem from a lack of factual accuracy or a failure to faithfully follow the provided input, further illustrating the multifaceted nature of hallucinations. Beyond this binary classification, hallucinations can also be understood in relation to knowledge grounding and verification. In knowledge-intensive tasks, such as open-domain question answering, the lack of explicit external evidence often leads to hallucinations, making it difficult to verify the correctness of generated answers. This further increases the risk of producing seemingly reasonable but factually incorrect outputs [13].

In summary, hallucination in language generation models large language models (LLMs) is a multifaceted phenomenon stemming from limitations in language generation and the knowledge grounding base. This complexity necessitates robust methods for detecting, estimating, and mitigating hallucination, which will be explored in the following chapters.

2.2 Causes of Hallucination

Hallucinations in large language models (LLMs) arise from a combination of data-related, model-related, and task-specific factors. These factors interact in complex ways, contributing to the generation of outputs that are fluent but factually incorrect.

From a data perspective, source-reference divergence in training data is a significant cause of hallucinations. When the target text contains information not supported by the input, the model learns generation patterns that do not fully rely on the source text [4]. This bias often stems from heuristic data collection processes or inherent characteristics of certain natural language generation tasks. Furthermore, large-scale training corpora may contain noise, biases, duplicate content, or unverified information. Consequently, the model may internalize and reproduce these defects, leading to inaccurate or unsubstantiated outputs [4]. These data-related issues collectively contribute to the generation of hallucinatory content in natural language models. More broadly, they reflect underlying data quality challenges, where inconsistencies between source and target, as well as noisy or biased training corpora, can systematically influence model behaviour.

From a modeling perspective, Large Language Models (LLMs) typically suffer from poor calibration and are prone to overconfidence, assigning excessively high confidence levels to incorrect predictions [14]. This phenomenon is closely related to the model's autoregressive training objective. LLMs primarily generate output by predicting the next most likely token, which makes the model more inclined during training to generate fluent, coherent, and confident outputs, rather than rigorously ensuring factual accuracy or estimating inherent uncertainty [15].

From a task and environment perspective, hallucinations are particularly prevalent in open-domain and knowledge-intensive settings. In these scenarios, he model primarily relies on parametric knowledge learned during pre-training and cannot acquire explicitly provided supporting evidence. The lack of reliable external evidence makes it extremely difficult to verify the correctness of the generated answers, which significantly increases the risk of producing seemingly reasonable but actually wrong answers.[16]. This challenge is even more pronounced in long-form content generation tasks, as maintaining factual consistency becomes increasingly difficult with longer outputs [17].

Overall, these factors not only explain the prevalence of hallucination, but also highlight the limitations of existing approaches that rely on single-source verification or surface-level evaluation signals. Hallucinations in LLMs result from the combined influence of imperfect training data, limitations in model calibration and objectives, and the inherent challenges of knowledge-intensive tasks. These insights motivate the need for more robust and integrated approaches to effectively detect, estimate, and mitigate hallucinations, as discussed in the following sections.

2.3 Hallucination Detection Methods

A wide range of approaches has been proposed to detect hallucinations in large language models (LLMs). Existing methods can be broadly categorised based on the source of information used for evaluation, including internal signal-based methods, external verification-based methods, and model-based evaluation methods. Internal signal-based methods rely on the model's own uncertainty or output variability, such as uncertainty-based and consistency-based approaches. External verification-based methods assess factual alignment using external evidence, including entailment-based and retrieval-based approaches. In addition, model-based evaluation methods utilise large language models as evaluators, commonly referred to as LLM-as-a-judge methods. While these approaches differ in their underlying assumptions, required resources, and robustness across tasks, each captures only partial aspects of hallucination. This section reviews these methods and provides a critical analysis of their strengths and limitations.

2.3.1 Internal signal-based methods

Uncertainty-based methods

Uncertainty-based methods detect hallucinations by estimating the model's uncertainty in its generated outputs. A common approach is to analyse internal probability signals, such as token-level probabilities, conditional sequence probabilities (log-likelihood), or entropy measures, under the assumption that higher uncertainty (or lower confidence) is associated with an increased risk of unreliable or hallucinated content [18].

In practice, these methods are widely adopted due to their simplicity and computational efficiency. They are typically unsupervised and training-free, relying solely on the model's internal output without requiring additional models, external knowledge bases, or retrieval mechanisms. Therefore, they provide a lightweight and practical solution for hallucination detection, especially suitable for resource-constrained environments [18].

However, uncertainty estimation based solely on model confidence is fundamentally limited by the poor calibration of large language models. Prior research has shown that LLMs often exhibit overconfidence, producing plausible but incorrect outputs with high probability [19]. As a result, internal confidence signals cannot reliably reflect factual correctness, making overconfident hallucinations difficult to detect. Furthermore, these methods do not incorporate external evidence or semantic consistency, limiting their effectiveness in complex or knowledge-intensive tasks.

In this work, uncertainty-based signals are not explicitly used. Instead, we focus on a multi-signal evaluation framework that combines semantic consistency (SelfCheckGPT), NLI-based factuality, and LLM-as-a-judge. This design addresses the limitations of confidence-based methods by introducing external and cross-sample validation signals, enabling more reliable hallucination detection in scenarios where model confidence is poorly calibrated.

Consistency-based methods

Consistency-based methods detect hallucinations by analyzing the agreement among multiple sampled responses generated for the same input. The underlying assumption is that when a model possesses correct knowledge, different sampled outputs tend to be consistent, whereas hallucinated content is more likely to diverge or contradict across samples [10].

In practice, these methods generate multiple candidate responses through stochastic sampling and evaluate their consistency using techniques such as semantic similarity, contradiction detection, natural language inference (NLI), or answer aggregation. For example, SelfCheckGPT [10] estimates hallucination risk by identifying contradictions among sampled responses without relying on external knowledge sources. Similarly, self-consistency [20] improves reasoning performance by selecting the most consistent answer across multiple sampled reasoning paths.

These approaches provide a lightweight and training-free mechanism for detecting hallucinations, enabling semantic validation through cross-sample agreement without requiring additional supervision or retrieval.

However, consistency-based methods also have several limitations. First, they require multiple sampling and forward passes, increasing computational cost and inference latency [4]. Second, they assume that correct answers are consistent across samples, which may not hold when multiple valid answers exist [21]. More importantly, consistency does not guarantee factual correctness, as models may produce highly consistent but incorrect responses due to systematic biases or flawed parametric knowledge [22].

In this work, consistency-based signals are incorporated as one component of a multi-signal hallucination detection framework. Specifically, SelfCheckGPT is used to capture cross-sample consistency, but is complemented by NLI-based factuality and LLM-as-a-judge evaluation. This design addresses the key limitation of pure consistency methods—namely, their inability to ensure factual correctness—by integrating external semantic validation signals and enabling more robust hallucination detection.

2.3.2 External / verification-based methods

Entailment-based methods (NLI)

Entailment-based methods use Natural Language Inference (NLI) to verify whether generated content is logically supported by references or evidence, thereby detecting hallucinations. The core idea is to model hallucination detection as a textual entailment problem: if the generated answer is implied by the reference text, it is considered reliable; if it contradicts the reference or lacks supporting evidence, it is considered a hallucination [23].

In practice, these methods typically employ pre-trained NLI models (e.g., models trained on MultiNLI) to compute entailment, contradiction, and neutral probabilities between reference text and generated answers [24]. Entailment scores are used to measure factual consistency, while contradiction signals help identify and penalise incorrect or unsupported content [25]. Prior work has shown that NLI-based approaches can effectively detect factual inconsistencies across tasks such as summarization and question answering [26].

A key advantage of entailment-based methods is their ability to capture semantic relationships beyond surface-level matching, enabling robust detection of factual inconsistencies. Compared to uncertainty-based or consistency-based methods, NLI explicitly evaluates the alignment between generated content and reference knowledge, making it particularly suitable for knowledge-intensive tasks.

However, these methods also have limitations. Their effectiveness depends on the quality and coverage of reference evidence, which may be incomplete or noisy in real-world scenarios [23]. In addition, NLI models themselves are imperfect and may produce erroneous judgments in complex contexts.

In this project, entailment-based signals play a central role in the hallucination detection framework. Unlike standard NLI-based approaches, we extend this paradigm by incorporating both positive (correct-answer entailment) and negative (incorrect-answer entailment and contradiction) evidence, enabling more fine-grained discrimination of factual correctness. Furthermore, NLI signals are combined with consistency-based evaluation (SelfCheckGPT) and LLM-as-a-judge scoring to form a multi-signal detector, mitigating the limitations of relying on NLI alone and improving overall robustness.

Retrieval-based methods

Retrieval-based methods aim to reduce hallucinations by grounding model outputs in external knowledge sources, such as documents or knowledge bases. By retrieving relevant evidence and verifying generated content against it, these methods improve factual reliability and reduce reliance on parametric knowledge [9].

In practice, retrieval-based approaches typically combine a retrieval module (e.g., BM25 or dense retrievers) with a generation or verification model, either by incorporating retrieved evidence during generation (e.g., Retrieval-Augmented Generation) or by performing post-hoc verification [27].

A key advantage of these methods is their ability to incorporate up-to-date external knowledge and provide explicit evidence for model outputs, thereby improving both factual accuracy and interpretability.

However, their effectiveness is highly dependent on the quality of the retrieval component. Missing or irrelevant evidence can lead to incorrect predictions, while multi-stage retrieval pipelines introduce additional system complexity and computational cost [9].

In this work, we do not employ retrieval-based grounding. Instead, we focus on a fully model-based, multi-signal detection and mitigation framework that operates without external knowledge sources. This design enables a fair comparison across datasets such as TruthfulQA, FEVER, and Natural Questions, where retrieval conditions are not uniformly available, and highlights the effectiveness of detector-guided correction without reliance on external evidence.

LLM-as-a-Judge methods

LLM-as-a-Judge methods utilize large language models as evaluators to assess the factual correctness of generated content. Instead of relying solely on internal confidence signals or external retrieval, these approaches reformulate hallucination detection as a reasoning-based evaluation task, where the model explicitly determines whether a given answer is truthful and semantically consistent with the reference [28].

In practice, evaluation is guided through carefully designed prompts that incorporate the input question, generated answer, and optionally reference answers or evidence. The model then produces discrete judgments (e.g., true/false) or scalar scores reflecting factual correctness [29]. This allows the evaluator to leverage its implicit knowledge and reasoning capabilities, enabling flexible and task-agnostic assessment across diverse settings.

A key advantage of LLM-as-a-Judge methods is their ability to capture semantic correctness beyond surface-level matching. Unlike uncertainty-based approaches, they do not depend on internal probability estimates, and unlike retrieval-based methods, they do not require access to external knowledge sources. As a result, they are particularly suitable for open-ended tasks such as TruthfulQA, where semantic correctness and misconception rejection are critical.

However, these methods also exhibit important limitations. First, the reliability of the evaluation depends on the capability and calibration of the evaluation model itself, which may still be biased or prone to hallucinations [30]. Second, evaluation results can be sensitive to prompt design and exhibit instability under different prompting or sampling conditions [31]. Third, LLM-based evaluation introduces additional computational overhead, especially when multiple candidate answers are assessed.

In this work, we address these limitations by integrating LLM-as-a-Judge as one component within a multi-signal hallucination detector, rather than relying on it as a standalone decision mechanism. Specifically, the judge output is combined with NLI-based factuality signals and self-consistency measures, and further calibrated using a learned scoring function. This design improves robustness against individual signal bias and enables more stable and reliable hallucination detection across datasets.

Furthermore, the judge signal is not only used for detection but also plays a role in the mitigation pipeline, where it contributes to candidate filtering and reranking. This integration allows the system to leverage semantic evaluation while mitigating its inherent instability through multi-signal aggregation and calibration.

2.4 Hallucination Mitigation Methods

Hallucination mitigation methods aim to reduce the generation of factually incorrect or unsupported content by large language models, complementing detection approaches by actively improving output reliability. Unlike detection methods, which identify hallucinations after generation, mitigation methods focus on preventing or correcting such errors during or after the generation process.

2.4.1 Prompt-based approaches

Prompt-based mitigation methods aim to reduce hallucinations by influencing the model's generative behaviour through carefully designed input instructions. These approaches do not modify model parameters but instead guide the model to produce more cautious, structured, and evidence-aware responses. Prior work has shown that well-designed prompts, including few-shot examples and instruction tuning, can significantly improve performance without additional training [32].

In particular, prompting strategies that encourage uncertainty awareness, such as allowing the model to abstain or avoid unsupported claims, have been shown to reduce hallucinations by promoting conservative generation [8]. Similarly, reasoning-oriented prompting techniques, such as Chain-of-Thought (CoT), guide the model to decompose complex problems into intermediate steps, thereby improving factual consistency and reasoning accuracy across a range of tasks [33].

Despite their effectiveness, prompt-based methods are inherently heuristic and highly sensitive to prompt design. The same prompt may yield significantly different performance across models and tasks, and even small variations in prompt wording can lead to unstable outputs [31]. Moreover, such methods do not fundamentally address the underlying causes of hallucination, as generation remains dependent on the model's internal parametric knowledge [21]. As a result, prompt-based techniques alone cannot guarantee factual correctness, especially in knowledge-intensive or open-domain settings.

These limitations motivate more structured and adaptive prompting strategies that integrate external evaluation signals into the generation process. Instead of relying on static, one-shot prompts, recent approaches explore multi-stage and feedback-driven prompting mechanisms, where model outputs are iteratively refined or selectively corrected. In this work, this idea is further extended by combining prompt-based correction with a detector-guided framework, where different prompting strategies (e.g., mild editing, strict correction, and abstention) are dynamically applied based on the estimated hallucination risk. This enables more targeted and reliable mitigation compared to conventional prompt-based methods.

2.4.2 Retrieval-augmented approaches

Retrieval-augmented approaches aim to mitigate hallucinations by grounding model outputs in external knowledge sources, such as documents, knowledge bases, or online corpora. Instead of relying solely on parametric knowledge, these methods incorporate retrieved evidence to improve factual accuracy and reliability [9].
In practice, retrieval-augmented methods typically consist of two main stages: (1) retrieving relevant documents using sparse or dense retrieval techniques (e.g., BM25 or neural retrievers such as DPR), and (2) conditioning the generation process on the retrieved context to produce evidence-grounded outputs [27]. By explicitly integrating external information, these approaches significantly improve performance in knowledge-intensive tasks such as open-domain question answering and fact verification [34]. In addition, grounding model outputs in retrieved evidence enhances interpretability, as predictions can be traced back to supporting documents.

However, retrieval-augmented approaches also have several limitations. Their effectiveness is highly dependent on the quality and relevance of the retrieved evidence; missing or noisy retrieval results can lead to incorrect or misleading outputs [4], [9], [27]. Furthermore, retrieval systems may introduce irrelevant information that interferes with reasoning and increases the risk of inconsistent predictions. These approaches also involve multi-stage pipelines (retrieval, ranking, and generation), which increase system complexity and computational cost.

These limitations motivate alternative approaches that do not rely on external retrieval. In this work, we adopt a training-free, model-based framework that operates without access to external knowledge sources. Instead, hallucination detection and mitigation are achieved through a combination of internal consistency signals (SelfCheckGPT), NLI-based factuality evaluation, and LLM-as-a-Judge. This design enables effective hallucination mitigation in closed-book and resource-constrained settings, while avoiding the dependency on retrieval quality and external knowledge availability.

2.4.3 Post-hoc correction methods

Post-hoc correction methods aim to reduce hallucinations by iteratively improving or correcting the model output after the initial generation. Unlike prompt-based or retrieval-augmented methods, these methods do not directly intervene in the generation process but introduce additional feedback and editing steps to improve the quality and factual consistency of the generated response. For example, Self-Refine [35] uses the same language model to first generate an initial output, then produce feedback on that output, and iteratively refine it, significantly improving output quality without additional training. In practice, post-hoc correction methods typically adopt a multi-stage process: the language model first generates an initial response, which is then iteratively improved through evaluation and modification steps. A mainstream strategy is self-correction, which prompts the same model to re-evaluate its output and identify potential factual errors, inconsistencies, or logical issues. This process often combines external feedback signals (e.g., task success metrics or unit test results) for consistency checks with internal reasoning to generate reflective feedback, ultimately enabling explicit verification and correction of the output [10], [35], [36]. Existing research shows that iterative improvement and self-correction mechanisms can help models detect and correct their own errors, thereby significantly improving the factual reliability of the output. In particular, Self-Refine [35] employs self-feedback and iterative refinement, while SelfCheckGPT [10] uses multi-sampling consistency checks and internal feedback loops to identify inconsistencies and reduce hallucinations. These methods are especially effective when external knowledge is limited or lacking, as they primarily rely on the model's own internal reasoning ability and multi-round self-assessment process.

However, post-hoc correction methods also have several important limitations. First, their effectiveness is essentially limited by the capabilities of the underlying model, as both generation and error correction depend on the same or similar model representations and reasoning patterns, which can easily lead to the retention or even reinforcement of systematic errors [35], [36]. In addition, such methods usually incur high computational overhead, and the quality of self-feedback is difficult to guarantee when external knowledge is absent or the base model is weak [10]. These limitations highlight the need to develop a more robust multi-signal detection and intervention framework.

Overall, post-hoc correction methods provide a flexible and model-agnostic approach to improving factual consistency by introducing an additional verification stage after generation. However, their reliance on internal model capabilities and iterative refinement may limit their effectiveness, particularly when the base model exhibits systematic errors or lacks sufficient knowledge.

In contrast, this work extends the idea of post-hoc correction by incorporating a multi-signal hallucination detector to guide the correction process. Instead of relying solely on self-feedback, the proposed framework integrates NLI-based factuality signals, LLM-as-a-judge evaluation, and self-consistency to evaluate and rerank multiple candidate answers. This detector-conditioned correction mechanism enables more reliable identification and mitigation of hallucinations, addressing the limitations of purely self-refinement-based approaches.

2.4.4 Training-based methods (RLHF / fine-tuning)

Training-based methods aim to mitigate hallucinations by directly modifying model parameters through additional training procedures, such as supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF). These approaches improve alignment and response quality by training on curated datasets or optimizing model behavior with human preference signals [32]. As a result, they can enhance the overall reliability and helpfulness of model outputs across a wide range of tasks.

However, such methods have several limitations. First, alignment does not guarantee factual correctness, and models may still produce plausible but incorrect responses, a phenomenon known as "alignment without truthfulness" [32]. Second, these approaches rely heavily on high-quality annotated data and incur substantial computational and human annotation costs [37]. Finally, they require access to model parameters, making them less flexible for black-box or deployment settings.

In contrast, this work focuses on a training-free, post-hoc framework that operates without modifying model parameters. Instead, hallucination detection and mitigation are achieved through multi-signal evaluation, including NLI-based factuality, LLM-as-a-judge, and self-consistency. This design enables effective hallucination mitigation in a plug-and-play manner, making it more practical for real-world applications where model retraining is not feasible.

2.5 Reproducibility Challenges in Hallucination Research

For large language models (LLMs), reproducibility has emerged as a critical challenge in hallucination research. Although deterministic decoding settings (e.g., fixed temperature and random seed) are commonly employed, the inherent stochasticity of the underlying generation process can still lead to significant output variability. Consequently, repeated runs of the same model on identical inputs may produce different responses, resulting in inconsistency in hallucination detection and evaluation [10]. A primary source of this variability lies in the probabilistic nature of language generation. Even under constrained decoding conditions, the outputs of Large Language Models (LLMs) are sampled from a probability distribution over possible continuations. Consequently, minor perturbations, such as changes in the sampling seed, adjustments to decoding parameters, or even subtle fluctuations in the underlying computational environment, can lead to significant discrepancies in both the generated content and downstream evaluation results [31]. This inherent uncertainty arising from probabilistic sampling contributes to the model's diminished robustness, particularly when processing edge cases or out-of-distribution inputs.

Furthermore, hallucination detection methods often exhibit instability. Prompt-based evaluation approaches are highly sensitive to prompt engineering, where minor changes in wording or format can significantly alter the model's judgment. Moreover, LLM-as-a-judge methods are frequently affected by position bias, verbosity bias (favoring longer answers), and scoring inconsistencies, further weakening the reliability of the evaluation results [28]. These issues are exacerbated by cross-dataset variability: detection algorithms that perform well on evidence-retrieval tasks often struggle to generalize effectively to more adversarial or open-ended tasks, reflecting the limited robustness of current hallucination detection approaches across diverse evaluation settings.

Motivated by these limitations, this project incorporates a stability-aware evaluation framework that complements traditional accuracy metrics with statistical measures such as mean performance and variance across repeated runs. This enables a more robust and reproducible assessment of hallucination detection and mitigation performance across different datasets and experimental settings.

2.6 Summary

In summary, existing research has explored a wide range of approaches for hallucination detection and mitigation in large language models, including prompt-based methods, retrieval-augmented generation, post-hoc correction, and training-based alignment. While these methods have demonstrated effectiveness under specific conditions, they often exhibit limitations in robustness, stability, and generalization across tasks and datasets.

In particular, many approaches rely on single evaluation signals, and detection and mitigation are typically treated as separate processes. Moreover, as discussed in Section 2.5, hallucination evaluation itself is inherently unstable due to stochastic generation, prompt sensitivity, and model biases. These challenges are further exacerbated in cross-dataset scenarios, where performance varies significantly between evidence-grounded and open-domain tasks.

This review identifies three key limitations in existing work:

- Over-reliance on single-signal detection methods
- Lack of integration between detection and mitigation
- Limited robustness and reproducibility across datasets

To address these challenges, this work proposes a multi-signal, detector-guided hallucination mitigation framework that integrates semantic evaluation (LLM-as-a-Judge), entailment-based verification (NLI), consistency-based signals (SelfCheckGPT), and additional auxiliary signals.

By combining these complementary signals within a unified scoring and calibration framework, the system estimates hallucination risk and dynamically applies mitigation strategies. In particular, prompt-based correction is used as an intervention mechanism, where different prompts (e.g., mild editing, strict correction, and abstention) are adaptively selected based on the predicted risk level.

Together with stability-aware evaluation, the proposed approach aims to achieve more robust, generalizable, and reproducible hallucination detection and mitigation across diverse tasks.

Chapter 3

Methodology and System design

This chapter introduces the methodological framework and system design developed for hallucination detection, estimation and mitigation in large language models. Existing hallucination detection approaches typically rely on a single signal, such as model confidence, entailment scores, or self-consistency. However, these methods suffer from well-known limitations, including instability across prompts, sensitivity to model bias, and limited effectiveness in cases where outputs are logically coherent but factually incorrect.

In contrast, the proposed framework introduces a unified multi-signal and closed-loop solution that jointly models detection, estimation, and mitigation, enabling more robust and stable performance across diverse datasets.

Unlike prior work that treats hallucination detection as a purely post-hoc evaluation task, this framework tightly integrates detection with generation through a feedback-driven mechanism. Specifically, detection signals are used to guide subsequent mitigation and answer refinement, forming a closed-loop system that iteratively improves output reliability.

To achieve this, the framework combines multiple complementary signals to evaluate factual correctness, including semantic judgment (LLM-as-a-Judge), entailment-based verification (NLI), cross-sample consistency (SelfCheckGPT), and additional auxiliary signals. These signals are aggregated into a unified and calibrated scoring function, which is further used to trigger adaptive mitigation strategies, such as answer rewriting, candidate generation, filtering, reranking, and abstention.

A key feature of the proposed approach is the detector-conditioned mitigation mechanism, where the level of intervention is dynamically determined based on the estimated reliability of each answer. This enables selective and targeted correction, avoiding unnecessary modifications to already correct outputs.

While the overall architecture is general, the system incorporates lightweight dataset-specific adaptations (e.g., prompt design, parsing strategies, and output constraints) to accommodate different task formulations without requiring model retraining.

In addition, this chapter presents the threshold selection and calibration procedures used to convert detection scores into actionable decisions, as well as the evaluation protocols designed to ensure robustness and reproducibility of experimental results.

Overall, the proposed design is motivated by the observation that hallucination is a multi-faceted phenomenon involving semantic inconsistency, factual contradiction, and stochastic instability. Therefore, integrating multiple complementary signals within a unified and feedback-driven framework is essential for achieving robust and generalisable hallucination detection and mitigation.

3.1 Problem Formulation

This project addresses hallucination detection, estimation, and mitigation in large language models (LLMs) through a prompt-engineering-based and detector-guided framework. Given an input query, which may take the form of a question or a claim, optional supporting evidence or context depending on the task setting, and a model-generated answer, the objective is to assess the factual reliability of the answer and, when necessary, apply corrective intervention to improve answer quality.

Formally, let q denote the input question or claim, $\tilde{e}^{(d)}$ the task-dependent evidence or context, and a the answer generated by the language model. The goal is to compute a dataset-dependent reliability score $s^{(d)}$ that reflects the degree of factual correctness of a. This score is defined as:

$$
\begin{aligned}
s^{(d)} = \mathcal{C}^{(d)}\Big(& w_1^{(d)} f_{\text{judge}}(q, a) \\
& + w_2^{(d)} f_{\text{NLI}}^{(d)}(\tilde{e}^{(d)}, a) \\
& + w_3^{(d)} f_{\text{consistency}}(\{a_i\}_{i=1}^{N}) \\
& + w_4^{(d)} f_{\text{lexical}}^{(d)}(a, r) \\
& - \lambda^{(d)} \mathbf{1}_{\text{contra}}^{(d)} f_{\text{contra}}(\tilde{e}^{(d)}, a) \Big)
\end{aligned}
\tag{1}
$$

where

$$
\tilde{e}^{(d)} = \begin{cases} e, & \text{if explicit evidence is available} \\ c(q), & \text{if context is constructed from source documents} \\ \varnothing, & \text{otherwise} \end{cases}
\tag{2}
$$

Here, f_{judge} denotes a semantic evaluation signal, $f_{\text{NLI}}^{(d)}$ denotes a dataset-dependent entailment or semantic consistency signal, $f_{\text{consistency}}$ measures agreement across multiple sampled answers, $f_{\text{lexical}}^{(d)}$ captures lexical overlap with a reference answer, and f_{contra} represents a contradiction-sensitive penalty.

Each component in Eq. 1 can be further defined as:

$$
f_{\text{judge}}(q, a) = \begin{cases} 1, & \text{if the judge evaluates the answer as correct} \\ 0, & \text{otherwise} \end{cases}
\tag{3}
$$

$$
f_{\text{consistency}} = 1 - \frac{1}{N} \sum_{i=1}^{N} \mathbb{P}_{\text{NLI}}(\text{contradiction} \mid a, a_i)
\tag{4}
$$

$$
f_{\text{lexical}}^{(d)}(a, r) = \text{TokenF1}(a, r)
\tag{5}
$$

$$
f_{\text{contra}}(\tilde{e}^{(d)}, a) = \mathbb{P}_{\text{NLI}}(\text{contradiction} \mid \tilde{e}^{(d)}, a)
\tag{6}
$$

The coefficients $w_i^{(d)}$ and $\lambda^{(d)}$ are dataset-dependent weights. The function $\mathcal{C}^{(d)}(\cdot)$ denotes an optional calibration or normalization step that maps the aggregated score into a bounded reliability range (e.g., $[0, 1]$), enabling consistent interpretation across different datasets and signals.

In practice, this calibration function can be instantiated as a logistic transformation:

$$\mathcal{C}^{(d)}(x) = \frac{1}{1 + e^{-(\alpha^{(d)}x + \beta^{(d)})}} \tag{7}$$

This formulation defines a unified, dataset-aware scoring framework, where different signals are emphasized according to task characteristics. For example, evidence-grounded tasks place greater emphasis on entailment and contradiction signals, while extractive question answering relies more on lexical overlap and short-answer alignment. This design maintains a consistent scoring principle while enabling flexible adaptation across heterogeneous evaluation settings.

Different datasets (e.g., FEVER, TruthfulQA, and Natural Questions) exhibit heterogeneous supervision signals and answer formats. In evidence-grounded datasets such as FEVER, NLI-based entailment and contradiction signals play a central role. In contrast, TruthfulQA relies more on semantic alignment with reference answers, with contradiction penalties applied more conservatively. For span-based question answering tasks such as Natural Questions, lexical overlap becomes the dominant correctness signal, while NLI is used as an auxiliary consistency check.

The score $s^{(d)}$ is then used both as a continuous estimate of factual reliability and as the basis for binary hallucination detection. Given a dataset-dependent threshold $\tau^{(d)}$, the answer is classified as:

$$\hat{y} = \begin{cases} \text{reliable} & \text{if } s^{(d)}(q, a) \geq \tau^{(d)} \\ \text{hallucinated} & \text{otherwise} \end{cases} \tag{8}$$

This decision function provides the detection component of the framework. However, unlike conventional approaches that treat detection as a purely post-hoc classification task, this work further extends the formulation into a closed-loop setting in which the detection result directly guides mitigation.

Specifically, the detector-conditioned mitigation process is formulated as a function:

$$a' = g^{(d)}(q, \tilde{e}^{(d)}, a, s^{(d)}) \tag{9}$$

where a' denotes the final refined answer after mitigation. The function $g^{(d)}(\cdot)$ applies different intervention strategies depending on the estimated reliability score. This decision process can be further defined as:

$$a' = \begin{cases} a, & s^{(d)} \geq \tau_{\text{high}}^{(d)} \\ \text{Edit}(a), & \tau_{\text{low}}^{(d)} \leq s^{(d)} < \tau_{\text{high}}^{(d)} \\ \text{SelectBestCandidate}(\mathcal{A}), & s^{(d)} < \tau_{\text{low}}^{(d)} \end{cases} \tag{10}$$

Here, $\text{Edit}(a)$ denotes a mild correction operation that modifies potentially incorrect parts of the original answer while preserving its structure, and $\text{SelectBestCandidate}(\mathcal{A})$ denotes a candidate-based selection process, where a set of alternative answers $\mathcal{A}$ is generated and the most reliable one is selected based on the proposed multi-signal scoring mechanism.

Overall, this formulation defines hallucination handling as a unified score-driven process that jointly supports continuous estimation, binary detection, and adaptive mitigation. The detailed design of the scoring components, dataset-specific signal configurations, and mitigation procedures is presented in the following sections.

3.2 System Overview

This section presents an overview of the proposed framework for hallucination detection, estimation, and mitigation in large language models. As illustrated in Figure 1, the system is designed as a closed-loop pipeline that integrates answer generation, reliability estimation, and corrective intervention into a unified process. Unlike conventional approaches that treat hallucination detection as a post-hoc classification task, the proposed framework uses reliability estimation to actively guide the generation process.

Given an input claim or question q and an optional evidence context e (if available), the system first generates an initial answer a using a large language model. This answer is then analysed by a multi-signal detection module, which estimates factual reliability from complementary perspectives. These signals include semantic correctness, factual consistency, contradiction-aware signals, and cross-sample consistency, allowing the system to

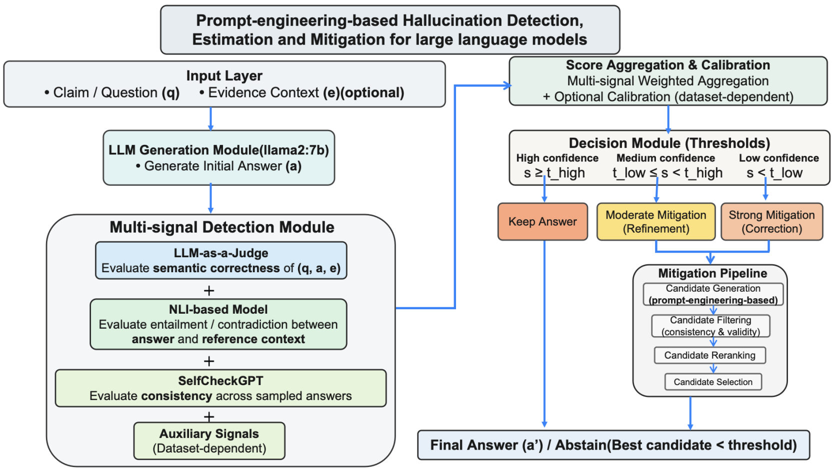

Figure 1: *Overview of the proposed prompt-engineering-based framework for hallucination detection, estimation, and mitigation. The system operates as a closed-loop pipeline that integrates answer generation, multi-signal reliability estimation, score aggregation, and threshold-based decision-making. Based on the estimated reliability, the system either accepts the initial answer or applies mitigation through candidate generation, filtering, and reranking.*

capture different types of hallucination behaviour.

The outputs of these signals are aggregated into a unified reliability score through a weighted combination and calibration process. This score provides a continuous estimate of factual correctness and serves as a control signal for downstream decision-making.

Based on the estimated reliability, a decision module determines whether the answer should be accepted or further refined. High-confidence answers are preserved, while lower-confidence answers trigger mitigation procedures of varying intensity. This results in a detector-conditioned intervention mechanism, where the level of correction is dynamically adapted based on the estimated hallucination risk.

When mitigation is required, the system generates multiple candidate answers and applies a structured selection process to identify the most reliable one. This process includes consistency-based filtering and reliability-aware reranking. The filtering criteria are dataset-aware and may rely on evidence when available (e.g., FEVER and Natural Questions), or on internal consistency signals when external grounding is not provided (e.g., TruthfulQA). If no candidate meets the required reliability criteria, the system may abstain from producing a final answer.

Overall, this design establishes a closed-loop framework in which hallucination detection, estimation and mitigation are tightly coupled. By using reliability estimation to guide corrective interventions, the system not only identifies hallucinated outputs but also actively improves them while maintaining answer quality. This integration distinguishes the proposed approach from conventional post-hoc detection methods, enabling a more robust and adaptive hallucination mitigation framework.

3.3 Datasets

To comprehensively evaluate the effectiveness, robustness, and generalisation ability of the proposed hallucination detection, estimation, and mitigation framework, three widely used benchmark datasets are employed: TruthfulQA, FEVER, and Natural Questions (NQ). These datasets are selected to represent diverse hallucination scenarios, ranging from adversarial questions to evidence-based verification and open-domain extractive QA.

3.3.1 TruthfulQA

TruthfulQA is designed to evaluate whether large language models produce truthful answers to questions that are prone to common misconceptions. Many questions are intentionally constructed such that commonly memorised or intuitive responses are incorrect. This dataset is particularly suitable for hallucination evaluation because:

- It explicitly targets model-generated misinformation
- It provides both correct and incorrect reference answers
- It requires semantic understanding rather than surface-level matching

In this work, TruthfulQA is used to evaluate hallucination behaviour in open-ended generative settings without explicit evidence, representing ungrounded scenarios.

3.3.2 FEVER

FEVER is a fact verification dataset consisting of claims labelled as *SUPPORTED*, *REFUTED*, or *NOT ENOUGH INFORMATION*, with corresponding evidence from Wikipedia. This dataset is suitable because:

- It provides explicit grounding through evidence

- It enables clear verification of factual consistency
- It supports structured evaluation via labels

In this project, FEVER represents fully grounded settings where hallucination can be evaluated against explicit evidence.

3.3.3 Natural Questions (NQ)

Natural Questions is an open-domain QA dataset consisting of real user queries with answers derived from Wikipedia documents, typically as extractive spans.
This dataset represents:

- Realistic user queries
- Context-dependent answers
- Noisy and long documents

In this work, NQ is used to evaluate hallucination in partially grounded settings, bridging ungrounded and fully grounded scenarios.

3.3.4 Summary

The combination of these datasets enables evaluation across:

- Ungrounded generation (TruthfulQA)
- Evidence-grounded verification (FEVER)
- Context-dependent extractive QA (NQ)

This design ensures that the proposed framework is evaluated across diverse hallucination settings and is not overfitted to a single task type.

3.4 Input Preparation and Answer Generation

This stage defines the initial interface between the language model and the proposed hallucination detection framework. The objective is to generate structured and analyzable candidate answers while maintaining compatibility with downstream multi-signal evaluation.

3.4.1 Context Construction

Context construction is applied in a dataset-aware manner to provide appropriate grounding signals when available.

For tasks with explicit supporting information, relevant contextual evidence is incorporated into the model input. In extractive question answering settings, a localised context representation is constructed to focus on relevant regions of the source document, reducing noise while preserving essential information. In evidence-grounded verification tasks, annotated evidence is directly used as input without additional transformation, ensuring consistency with the dataset structure. For ungrounded tasks, no external context is provided, allowing the system to evaluate hallucination behaviour under purely parametric knowledge conditions.

This design ensures that grounding is introduced only when appropriate, enabling controlled comparison across datasets with different levels of evidence availability.

3.4.2 Prompting Strategy

Prompt engineering is used as the primary mechanism to control model behaviour across heterogeneous tasks. Instead of modifying model parameters, task-specific prompts impose structural and semantic constraints on the generated outputs.

In extractive settings, prompts encourage the model to produce answers aligned with the provided context. In verification tasks, prompts guide the model to generate outputs consistent with the input claim and associated evidence. In open-ended settings, prompts are designed to elicit concise and direct answers without external grounding.

Across all datasets, structured output formats are enforced to ensure consistency and enable reliable downstream processing. This unified prompting strategy provides controlled generation behaviour while maintaining flexibility across task settings.

3.4.3 Answer Generation

Given an input query and optional context, the system generates one or more candidate answers using a large language model. The generation process is designed to balance stability and variability, ensuring that outputs are both reliable and informative for downstream evaluation.

To capture uncertainty in model behaviour, multiple candidate answers are produced for each input:

$$A = \{a_1, a_2, \ldots, a_N\}$$

This multi-sample generation enables the estimation of cross-sample consistency, which serves as an important signal for hallucination detection. The level of variability is controlled to ensure that generated answers remain task-consistent while still exposing potential inconsistencies.

3.4.4 Output Normalisation

Raw model outputs are transformed into a standardised representation to ensure comparability across samples. This process includes structured extraction, format normalisation, and removal of non-essential content.

The normalisation strategy is dataset-aware. In open-ended tasks, it focuses on isolating concise answer content. In classification tasks, it ensures consistent mapping to predefined label spaces. In extractive settings, it enforces alignment with expected answer formats. This stage provides a clean and stable input for subsequent evaluation and scoring.

3.4.5 Validity Handling

To ensure robustness, outputs that do not satisfy structural or minimal content requirements are handled through a lightweight validation mechanism. Instead of aggressive correction, invalid outputs are normalised or replaced using conservative fallback strategies that preserve task constraints. This design maintains evaluation consistency while avoiding unintended semantic modifications.

3.4.6 Design Rationale

The answer generation module is designed to support the downstream hallucination detection framework rather than operate as an independent component. Specifically:

- Context-aware input construction enables controlled grounding
- Prompt-based control ensures consistent output structure
- Multi-sample generation supports consistency-based evaluation
- Output normalisation ensures compatibility with scoring modules

By aligning generation with detection requirements, the system enables reliable application of multi-signal hallucination evaluation and mitigation.

3.4.7 Summary

Overall, this module provides a structured and dataset-aware generation process that produces consistent and analyzable candidate answers. By combining controlled prompting, context-aware input design, and multi-sample generation, it establishes a robust foundation for subsequent hallucination detection and mitigation.

3.5 Hallucination Detection Model

This section presents the proposed hallucination detection model, which formulates hallucination detection as a multi-signal factual reliability estimation problem. Rather than relying on a single confidence measure, the model integrates multiple complementary signals to capture different aspects of hallucination behaviour, including semantic correctness, factual consistency, and response stability. This design improves robustness across heterogeneous tasks and reduces sensitivity to individual signal limitations.

The overall framework produces a continuous reliability score for each generated answer, which is subsequently used for both hallucination detection and downstream mitigation. Unlike conventional approaches that treat detection as an isolated post-hoc classification task, the proposed model is designed to support a closed-loop pipeline, where detection outputs directly guide corrective interventions.

3.5.1 Problem Formulation

Hallucination detection is formulated as a reliability estimation problem over generated answers. Given an input query q, a model-generated answer a, and optional task-dependent reference information r (e.g., evidence, context, or reference answers), the objective is to estimate the factual reliability of the answer.

Formally, the detection model defines a scoring function:

$$f^{(d)}(q, a, r) \to s^{(d)}(q, a) \in [0, 1] \tag{11}$$

where s represents the estimated reliability of the answer. Higher values indicate greater confidence in factual correctness, while lower values indicate a higher likelihood of hallucination.

A dataset-dependent threshold $\tau^{(d)}$ is applied to obtain a binary decision:

$$\hat{y} = \begin{cases} \text{reliable} & \text{if } s \geq \tau \\ \text{hallucinated} & \text{otherwise} \end{cases} \tag{12}$$

This formulation is designed to generalise across datasets with heterogeneous structures. Due to differences in supervision signals and task formulations, hallucination is operationalised differently across datasets. For example, extractive QA tasks emphasise lexical alignment, evidence-based verification tasks emphasise consistency with external evidence, and open-ended QA tasks emphasise semantic correctness relative to reference answers.

This motivates a unified yet flexible framework that adapts to dataset characteristics while maintaining a consistent underlying scoring principle.

3.5.2 Multi-signal Detection Framework

To improve robustness across diverse tasks, the proposed detector combines multiple complementary signals rather than relying on a single metric. Each signal captures a distinct aspect of factual reliability, and their integration enables a more comprehensive assessment.

Given an input (q, a, r), the framework constructs a multi-dimensional signal representation:

$$\mathbf{x}(q, a, r) = [x_1, x_2, \ldots, x_k] \tag{13}$$

where each component corresponds to a specific reliability signal, such as semantic evaluation, inference-based consistency, or cross-sample agreement.

The final reliability score is computed using an aggregation function:

$$s^{(d)}(q, a) = h^{(d)}(\mathbf{x}(q, a, r)) \tag{14}$$

where $h^{(d)}(\cdot)$ denotes a dataset-aware combination function.

This design allows the detector to remain general while adapting to different task settings, where the relative importance of signals may vary depending on the availability of evidence, reference answers, or contextual constraints.

3.5.3 Individual Signals

The proposed framework incorporates three primary signals, each capturing a complementary aspect of factual reliability.

Semantic Evaluation Signal The first signal provides a high-level semantic assessment of answer correctness. Given the input query and generated answer, this signal evaluates whether the answer is consistent with expected knowledge or task-specific references.

Formally, this signal can be represented as:

$$s_{\text{judge}} \in [0, 1] \tag{15}$$

where higher values indicate stronger semantic alignment with correct information.

Inference-based Factual Consistency The second signal evaluates the logical consistency between the generated answer and available reference information (e.g., evidence or context). This is achieved by estimating the degree of entailment and contradiction between textual inputs.

Formally, the signal is defined as:

$$s_{\text{fact}} = \max\left(p_{\text{ent}} - \alpha \cdot p_{\text{con}}, 0\right) \tag{16}$$

where p_{ent} and p_{con} denote entailment and contradiction scores, respectively, and α controls the penalty for contradiction.

This formulation explicitly penalises contradictory outputs, making it particularly effective for detecting hallucinations that conflict with available evidence.

Self-consistency Signal The third signal captures the stability of model outputs under stochastic generation. For each query, multiple candidate answers are generated, and their agreement is evaluated.

Given a set of sampled answers $\{a_1, \ldots, a_n\}$, the consistency signal is defined as:

$$s_{\text{sc}} = 1 - \frac{1}{n}\sum_{i=1}^{n} \text{Inconsistency}(a_i, \{a_j\}_{j\neq i}) \tag{17}$$

Higher values indicate stronger agreement across samples, suggesting greater reliability. This signal reflects model uncertainty, as hallucinated answers tend to exhibit higher variability.

3.5.4 Score Aggregation and Calibration

The individual signals are combined to produce a unified reliability score. Given the signal vector $\mathbf{x} = [s_{\text{judge}}, s_{\text{fact}}, s_{\text{sc}}]$, the aggregated score is computed as:

$$S^{(d)}(q, a) = w_j s_{\text{judge}} + w_f s_{\text{fact}} + w_s s_{\text{sc}} \tag{18}$$

where w_j, w_f, w_s are non-negative weights reflecting the relative importance of each signal.

To ensure comparability across datasets and improve interpretability, a calibration function is applied to map the aggregated score into a bounded reliability range:

$$s^{(d)}(q,a) = \mathcal{C}^{(d)}\big(S^{(d)}(q,a)\big) \tag{19}$$

where $S^{(d)}(q,a)$ denotes the aggregated multi-signal score, and $\mathcal{C}^{(d)}(\cdot)$ is a calibration function that maps the score into a bounded reliability range.

This calibration step produces a normalized reliability estimate that can be consistently interpreted across different tasks and experimental settings.

3.5.5 Auxiliary Mechanisms

In addition to the primary signals, several auxiliary mechanisms are incorporated to enhance robustness.
First, a contradiction-aware filtering mechanism suppresses outputs that exhibit strong factual inconsistency with available references.

Second, structural validity constraints are applied to ensure that candidate answers satisfy task-specific requirements, improving overall reliability.

Third, agreement-based reranking prioritises answers that are both factually consistent and stable across multiple candidates.

Finally, an abstention-aware mechanism allows the system to avoid producing unreliable answers when confidence is insufficient, thereby reducing the risk of hallucination.

These mechanisms complement the core scoring framework by enforcing additional constraints on answer validity and consistency.

3.5.6 Dataset-specific Adaptation

Although the framework adopts a unified multi-signal design, different datasets require task-specific adaptations due to variations in supervision signals and answer formats.

In evidence-grounded tasks, factual consistency and contradiction signals are emphasised to ensure alignment with external evidence. In open-ended tasks, semantic evaluation and self-consistency play a more prominent role in capturing correctness and stability. In extractive QA settings, lexical alignment becomes the dominant signal, ensuring that answers remain grounded in the provided context.

By adjusting the relative importance of signals according to task characteristics, the framework maintains a consistent scoring principle while remaining adaptable to heterogeneous evaluation settings.

Summary Overall, the proposed hallucination detection model defines a unified, multi-signal framework for estimating factual reliability. By integrating semantic evaluation, inference-based consistency, and self-consistency signals, and combining them through a calibrated scoring mechanism, the framework provides a robust and generalisable solution for hallucination detection across diverse tasks. This design also enables seamless integration with downstream mitigation, forming a complete closed-loop hallucination handling system.

3.6 Threshold-based Decision Mechanism

To convert the continuous reliability score into discrete hallucination predictions, a threshold-based decision mechanism is adopted. This step transforms the estimated reliability score into a binary classification, enabling direct identification of hallucinated and non-hallucinated outputs.

Formally, given a reliability score $s^{(d)}(q, a)$ for an answer a under dataset d, the prediction is defined as:

$$\hat{y} = \begin{cases} \text{reliable}, & s^{(d)}(q, a) \geq \tau^{(d)} \\ \text{hallucinated}, & s^{(d)}(q, a) < \tau^{(d)} \end{cases} \tag{20}$$

where τ denotes a decision threshold.

This formulation allows the continuous scoring function to be mapped to a discrete decision space, while preserving flexibility in how the reliability score is constructed. The threshold τ acts as a control parameter that balances sensitivity to hallucination and tolerance for uncertainty.

Although the underlying scoring function $s^{(d)}(q, a)$ varies across datasets due to differences in task formulation and supervision signals, the threshold-based decision mechanism remains consistent. In particular, different datasets emphasise different aspects of reliability: semantic correctness in open-ended tasks, evidence consistency in verification tasks, and lexical alignment in extractive question answering. Despite these differences, the same decision principle applies uniformly across settings.

This unified thresholding strategy ensures that the framework maintains a consistent interpretation of reliability while allowing dataset-specific variations in score construction.

3.6.1 Summary

Overall, the hallucination detection model can be viewed as a two-stage process. First, multiple complementary signals are extracted and aggregated into a unified reliability score $s^{(d)}(q, a)$. Second, a threshold-based decision rule is applied to convert this continuous estimate into a binary hallucination prediction.

This formulation provides a modular and extensible framework, where different signals and scoring strategies can be incorporated without altering the final decision mechanism. By decoupling score estimation from decision-making, the framework achieves both flexibility and consistency across heterogeneous tasks.

3.7 Mitigation Strategy

This section presents a closed-loop mitigation framework for reducing hallucinations in large language model outputs. Given an initial answer, the framework first estimates its factual reliability and then applies a detector-conditioned transformation to improve the answer or abstain when necessary.

Importantly, the same reliability function $s^{(d)}(q, a)$ is shared between the detection and mitigation stages, ensuring consistency across the closed-loop pipeline. This shared formulation ensures that both detection and mitigation are governed by a consistent objective, enabling principled optimisation over candidate answers.

The mitigation process is formulated as a decision problem over a candidate set. Given a question q and an initial answer a, the objective is to produce a refined answer a' that maximises factual reliability:

$$a' = \arg \max_{a \in \mathcal{C} \cup \{\text{abstain}\}} s^{(d)}(q, a) \tag{21}$$

where $\mathcal{C}$ denotes a set of candidate answers derived from a, and $s^{(d)}(q, a)$ is the reliability function.

This formulation highlights that both detection and mitigation are governed by the same reliability function, enabling a unified optimisation perspective over candidate answers.

3.7.1 Reliability-aware Decision Function

The mitigation process is guided by a reliability score:

$$s^{(d)}(q, a) \in [0, 1] \tag{22}$$

Based on this score, a detector-conditioned policy determines the level of intervention:

$$a' = \begin{cases} a, & s^{(d)}(q, a) \geq \tau_{\text{high}} \\ \text{Edit}(a), & \tau_{\text{low}} \leq s^{(d)}(q, a) < \tau_{\text{high}} \\ \text{SelectBestCandidate}(\mathcal{C}), & s^{(d)}(q, a) < \tau_{\text{low}} \end{cases} \tag{23}$$

This formulation enables adaptive mitigation, where stronger corrections are applied to lower-confidence answers.

3.7.2 Candidate Construction

For answers requiring intervention, a set of candidate responses $\mathcal{C}$ is constructed:

$$\mathcal{C} = \{a_1, a_2, \ldots, a_k\} \tag{24}$$

Candidate construction is performed through controlled transformations of the initial answer. These transformations aim to preserve correct information while improving factual reliability. Depending on the task setting, candidates may be generated through rewriting, refinement, or extraction from available context.

3.7.3 Candidate Selection

All candidates are evaluated using the same reliability function, ensuring consistency between detection and mitigation:

$$a^* = \arg\max_{a \in \mathcal{C}} s^{(d)}(q, a) \tag{25}$$

To improve robustness, candidates may be filtered based on validity and consistency constraints before selection.

3.7.4 Abstention Mechanism

To reduce the risk of incorrect outputs, an abstention option is incorporated. The final decision is extended as:

$$a' = \begin{cases} a^*, & s^{(d)}(q, a^*) \geq \tau_{\text{abs}}^{(d)} \\ \text{abstain}, & \text{otherwise} \end{cases} \tag{26}$$

This mechanism enables the system to avoid producing unreliable answers when confidence is insufficient.

3.7.5 Dataset-aware Adaptation

While the overall mitigation framework remains unified, its instantiation is adapted to different task settings. In open-ended generation, mitigation emphasises semantic correction and consistency. In evidence-grounded tasks, candidate selection is constrained by alignment with supporting information. In extractive settings, mitigation prioritises selecting context-grounded spans.

3.7.6 Summary

Overall, the proposed mitigation strategy formulates hallucination reduction as a score-driven optimisation process. By combining reliability estimation, adaptive intervention, and candidate-based selection, the framework provides a principled approach to improving factual correctness while controlling hallucination risk.

3.8 Summary

In this section, we presented a closed-loop framework that integrates answer generation, hallucination detection, estimation, and mitigation into a unified pipeline. The framework begins by producing candidate answers using a base language model, including both direct outputs and multiple sampled variations. A multi-signal hallucination detection module is then applied, combining complementary signals such as LLM-as-a-judge, NLI-based factual consistency, self-consistency, lexical overlap, and auxiliary constraints. These signals are aggregated into a continuous reliability score that quantifies the factual correctness of each answer.

Based on the estimated reliability, a detector-conditioned mitigation strategy is applied. This strategy dynamically determines the level of intervention, ranging from preserving the original answer to applying controlled correction or selecting alternative candidates through a structured filtering and reranking process.

The framework is further designed to be dataset-aware, allowing different signals and mitigation strategies to be emphasised depending on task characteristics. In evidence-grounded settings, the framework prioritises consistency with supporting information; in extractive tasks, it emphasises alignment with context spans; and in open-ended settings, it focuses on semantic correctness and consistency across generated responses.

Overall, this design provides a unified and modular approach to hallucination handling, where detection, estimation, and mitigation are tightly integrated through a score-driven decision process. This formulation enables adaptive intervention strategies while maintaining consistency across heterogeneous task settings.

Chapter 4

Implementation and Evaluation

This chapter presents the implementation and evaluation of the proposed hallucination detection, estimation, and mitigation framework.

The first part of this chapter focuses on the system implementation, describing how the framework is realised in practice, including pipeline execution, model integration, and engineering design choices. Emphasis is placed on practical considerations such as efficiency, reproducibility, and modularity, rather than conceptual design.

The second part of this chapter introduces the experimental setup used to evaluate the framework. This includes the datasets, model configurations, and evaluation metrics. Finally, the evaluation protocol is described, covering data splitting, threshold tuning, and Monte Carlo sampling procedures to ensure fair, robust, and reproducible results.

This structure ensures a clear separation between the methodological design and its practical implementation and evaluation.

4.1 Overall Pipeline

The system is implemented as a modular closed-loop pipeline, where answer generation, hallucination detection, and mitigation are executed sequentially within a unified workflow. Each component is implemented as an independent module, allowing flexible configuration and reuse across datasets.

Given an input query q, the system first generates one or more candidate answers using a base language model. These outputs are cached to ensure consistency and to avoid redundant computation during subsequent processing stages.

The hallucination detection module then evaluates each candidate answer using a multi-signal framework, producing a reliability score $s^{(d)}(q, a)$ for each candidate. This score serves as a unified interface between detection and mitigation.

Based on the estimated reliability, a mitigation module is applied to refine the answer. High-confidence answers are preserved, while lower-confidence answers trigger additional processing steps such as candidate generation, filtering, and reranking.

The pipeline is designed to be dataset-agnostic at the architectural level, while supporting lightweight task-specific adaptations. This ensures consistent evaluation across datasets while maintaining flexibility for different task requirements.

4.2 Answer Generation Implementation

The answer generation component is implemented as a reusable module shared across all datasets, with dataset-specific prompts and post-processing strategies. The implementation focuses on controlled sampling, reproducibility, and efficient reuse of generated outputs.

Generation Framework

Answer generation is performed using the `llama2:7b` model served via the Ollama framework. All model interactions are handled through a unified wrapper function with retry logic to improve robustness against transient failures. While the same generation backend is reused across datasets, decoding settings are configured in a dataset-specific manner.

For TruthfulQA, answer generation uses temperature 0.3 with top-p 0.9, encouraging short but slightly diverse responses. For FEVER, answer generation uses temperature 0.6 with top-p 0.9, allowing more variation in structured label-and-reason outputs. For Natural Questions, the base extractive answer generator uses temperature 0.0 to maximise determinism and promote short span-like answers aligned with the extractive setting.

Multi-Sample Generation

For each question, multiple candidate answers are generated to support downstream consistency-based signals. In the current implementation, the number of samples is fixed to $N_{\text{samples}} = 3$, ensuring that each question is associated with a consistent set of candidate answers. If cached answers already exist, they are reused to avoid redundant computation.

Answer Caching

To improve efficiency and reproducibility, generated answers are stored in a persistent CSV cache. Before generating new answers, the system checks whether sufficient samples are already available for a given question. If the required number of samples has already been generated, cached answers are reused directly. Otherwise, only the missing samples are generated and appended to the cache.

This design avoids redundant model calls, reduces computational cost, and ensures that all experiments operate on a consistent set of generated answers across multiple runs.

Dataset-Specific Prompting

Although the generation framework is shared, prompting strategies are adapted to each dataset:

- **TruthfulQA:** Answers are generated using a constrained prompt that enforces concise and direct responses. The model is explicitly instructed to produce a short answer in a fixed format ("`FINAL_ANSWER: <answer>`"). This design reduces verbosity and ensures a consistent answer structure for downstream parsing and scoring.

- **FEVER:** Prompt engineering is employed to enforce structured outputs that explicitly include label predictions and evidence-grounded explanations. The model is constrained to produce outputs in a fixed `LABEL + REASON` format, ensuring consistency with the claim verification task. This structured design enables reliable label extraction, facilitates NLI-based factuality assessment, and supports robust downstream evaluation and scoring.

- **Natural Questions:** Prompts are designed to produce short,extractive-style answers consistent with the dataset's evaluation criteria (exact match and token-level F1). This behavior is further reinforced through answer compression and span-based extraction mechanisms, ensuring that outputs remain concise and grounded in the provided context.

All prompts enforce a standard output format using a `FINAL_ANSWER` prefix, which simplifies downstream parsing and evaluation.

Post-processing

Generated outputs are normalised using a lightweight parsing function that extracts the final answer segment from model outputs. This step ensures consistent formatting across different generation modes and mitigates variability in model outputs.

Design Considerations

The generation module is designed to support downstream hallucination detection by providing diverse yet controlled candidate answers. Multi-sample generation enables the computation of peer consistency and self-consistency signals, while caching ensures computational efficiency and reproducibility. The use of dataset-specific prompts further aligns generation behaviour with task requirements, improving the reliability of subsequent detection and mitigation stages.

4.3 Hallucination Detection Implementation

4.3.1 TruthfulQA Detection Implementation

The detector is implemented as a multi-stage scoring pipeline that combines semantic evaluation, peer-based consistency, and self-consistency signals, followed by probabilistic calibration.

LLM-as-a-Judge

The LLM-as-a-Judge component is implemented using the `qwen2.5:7binstruct` model via the Ollama framework. The model operates in a *reference-free* setting and evaluates whether an answer appears truthful, cautious, and free from common misconceptions. A zero-temperature decoding strategy is used, and the output is converted into a binary score (1.0 for True, 0.0 for False).

Peer-based NLI Consistency

Semantic consistency across sampled answers is measured using the `roberta-large-mnli` model. For each candidate answer, entailment and contradiction scores are computed against other sampled answers for the same question. These scores are aggregated by averaging across all comparisons, producing a peer consistency signal:

$$\text{peer\_score} = \max(\text{entailment} - 0.7 \cdot \text{contradiction}, 0) \tag{27}$$

Self-Consistency

Self-consistency is computed using the SelfCheckGPT-NLI framework. Given multiple sampled answers, pairwise disagreement scores are calculated and averaged. The final self-consistency score is defined as:

$$\text{sc\_score} = 1 - \text{disagreement} \tag{28}$$

Heuristic Scoring

A heuristic reliability score is constructed as a weighted combination of multiple signals:

$$\begin{aligned} \text{heuristic\_score} = {} & w_1 \cdot \text{judge} \\ & + w_2 \cdot \text{peer\_score} \\ & + w_3 \cdot \text{sc\_score} \\ & + w_4 \cdot \max(\text{entailment} - \text{contradiction}, 0) \end{aligned} \tag{29}$$

In our implementation, the weights are defined as:

$$w_1 = 0.35, \quad w_2 = 0.20, \quad w_3 = 0.15, \quad w_4 = 0.20$$

where:

- judge is a binary signal (0 or 1) obtained from an LLM-as-a-Judge model,
- peer_score measures agreement with other sampled answers using NLI,
- sc_score $= 1 -$ SelfCheckGPT score reflects answer consistency,
- entailment and contradiction are derived from a MNLI model.

The term $\max(\text{entailment} - \text{contradiction}, 0)$ ensures that only positive factual support contributes to the score, while contradictory evidence is suppressed.

Design Rationale

The heuristic score is designed to combine complementary signals that capture different aspects of factual reliability.

First, the *judge* signal provides a high-level semantic assessment from a large language model, capturing global coherence and commonsense correctness. However, it may be overconfident and lacks fine-grained factual grounding.

Second, the *peer_score* reflects agreement between independently sampled answers. This signal captures answer stability and reduces the impact of stochastic generation, but cannot distinguish between consistently correct and consistently incorrect outputs.

Third, the *self-consistency score* (derived from SelfCheckGPT) measures internal agreement across generated samples. A high consistency indicates robustness, while low consistency often correlates with hallucination.

Finally, the MNLI-based term $\max(\text{entailment} - \text{contradiction}, 0)$ provides a token-level factuality signal. This term explicitly rewards entailment while suppressing contradictory evidence, ensuring that only positive factual support contributes to the score.

The combination of these signals enables the detector to leverage complementary strengths: semantic reasoning (judge), distributional agreement (peer and self-consistency), and textual entailment (MNLI).

This multi-signal design improves robustness compared to single-signal approaches, as different signals capture complementary aspects of hallucination, reducing the risk of systematic bias from any individual component.

Calibration and Final Scoring

The heuristic score is not intended to represent a calibrated probability. Instead, it serves as a structured feature aggregation capturing multiple reliability signals.

To obtain a well-calibrated estimate of hallucination risk, a logistic regression model is trained on validation data using these signals as input features. This corresponds to a form of Platt scaling, mapping the feature space to a probabilistic output p_{hall}.

The final score combines both components:

$$\text{score} = 0.7 \cdot (1 - p_{\text{hall}}) + 0.3 \cdot \text{heuristic\_score}$$

This design balances:

- **learned calibration** (via p_{hall}), which improves probabilistic accuracy and generalisation, and
- **interpretable structure** (via the heuristic score), which preserves signal transparency and stability.

Note that the heuristic weights do not sum to 1, as the score is not a probability distribution but a relative confidence measure that is further refined by calibration.

Abstention Handling

Answers identified as abstentions (e.g., "I do not know") are explicitly detected and down-weighted during scoring, reflecting their lower informational utility while still avoiding hallucinated content.

Ablation Settings

To analyse the contribution of each component, multiple detector variants are evaluated by selectively disabling individual signals (e.g., no-judge, no-NLI, no-self-consistency, judge-only), allowing us to quantify the impact of each signal on overall performance and stability.

Evaluation Protocol

All decision thresholds are tuned on a validation split and fixed during test-time evaluation, ensuring a fair and unbiased assessment.

Performance is measured using paired Monte Carlo sampling, which controls for variance in answer generation and enables robust comparison across detector configurations.

4.3.2 FEVER Detection Implementation

For the FEVER dataset, the detector is implemented in an evidence-grounded verification setting, where each instance consists of a claim, associated evidence, and a structured label (*SUPPORTED*, *REFUTED*, or *NOT ENOUGH INFO*). The detection objective is to assess whether a generated answer is both label-correct and logically consistent with the provided evidence.

LLM-as-a-Judge

A semantic evaluation signal is obtained using an LLM-based judge implemented with the `qwen2.5:7b-instruct` model via Ollama. The judge operates under a carefully designed evaluation prompt that verifies: (i) whether the predicted label matches the reference label, and (ii) whether the explanation avoids contradiction with the evidence. The output is a binary score $s_{\text{judge}} \in \{0, 1\}$.

NLI-based Factuality Signal

The primary factuality signal is computed using the `roberta-large-mnli` model. Given the evidence as premise and a label-derived hypothesis (e.g., "The claim is SUPPORTED"), the entailment (e) and contradiction (c) probabilities are obtained. To strongly penalise contradiction, the factuality score is defined as:

$$s_{\text{fact}} = \max(e - \alpha \cdot c, 0), \tag{30}$$

where $\alpha = 1.8$ is a tunable hyperparameter. It is empirically chosen to increase sensitivity to contradiction, reflecting the observation that contradictory evidence is a stronger indicator of hallucination than lack of entailment.

Self-Consistency

Self-consistency is computed using the SelfCheckGPT-NLI framework over multiple sampled answers. Each answer is converted into a label-based hypothesis (e.g., "The claim is REFUTED"), and pairwise contradiction scores are computed. The final self-consistency score is:

$$s_{\text{sc}} = 1 - \text{disagreement}. \tag{31}$$

Heuristic Scoring

The overall reliability score is computed as a weighted combination of the three signals:

$$\text{score} = w_1 \cdot s_{\text{judge}} + w_2 \cdot s_{\text{fact}} + w_3 \cdot s_{\text{sc}}, \tag{32}$$

where $w_1 = 0.35$, $w_2 = 0.45$, and $w_3 = 0.20$.

This weighted formulation emphasises factual verification (via s_{fact}) while still incorporating semantic reasoning and consistency signals.

GOLD NLI Contradiction Filtering

To further improve robustness and reduce model leakage, an auxiliary NLI model from a different family (DeBERTa-based) is used as a contradiction filter. Given the evidence and the predicted label hypothesis, the contradiction probability c_{gold} is computed. Candidates with strong contradiction are filtered out:

$$c_{\text{gold}} \geq \tau_{\text{gold}} \Rightarrow \text{reject candidate}, \tag{33}$$

where $\tau_{\text{gold}} = 0.50$.

Label-Constrained Evaluation

Unlike open-ended QA tasks, correctness in FEVER requires both:

- the predicted label to match the ground-truth label
- the answer to be non-contradictory with the evidence

This is enforced through a strict gold verification function that combines label matching and contradiction filtering. This makes FEVER detection fundamentally stricter than open-domain QA, as answers must be both semantically correct and explicitly supported by evidence.

Design Rationale

The FEVER detector is designed to operate under an evidence-grounded setting, where factual correctness depends not only on semantic plausibility but also on explicit consistency with supporting evidence.

The LLM-as-a-Judge component captures high-level reasoning about label correctness and explanation quality, but may be sensitive to prompt design and lacks strict factual grounding.

The NLI-based factuality signal provides a fine-grained verification mechanism, explicitly modelling entailment and contradiction between the evidence and the predicted claim label. The use of a contradiction-weighted formulation ensures that incorrect claims are strongly penalised.

Self-consistency captures agreement across multiple generated answers, improving robustness against stochastic generation effects, although it cannot distinguish between consistently correct and consistently incorrect outputs without additional grounding signals.

The additional GOLD NLI filtering step introduces an independent verification signal from a different model family, reducing potential model bias and preventing leakage between detection and evaluation components.

Together, these components provide complementary strengths, combining semantic reasoning, evidence-based verification, and robustness through agreement, which is particularly important in structured fact verification tasks such as FEVER. Compared to single-signal approaches, this multi-stage design improves robustness, as errors from individual components can be mitigated by complementary signals.

Ablation Settings

To analyse the contribution of each component, multiple detector variants are evaluated, including removing individual signals (e.g., no-judge, no-NLI, no-self-consistency) and a judge-only baseline.

Evaluation Protocol

All thresholds are tuned on a validation split and fixed for test evaluation. The detector is evaluated using paired Monte Carlo sampling across multiple runs to ensure robustness and fair comparison between configurations.

4.3.3 Natural Questions Detection Implementation

For Natural Questions, the detection component is implemented for a short-answer extractive setting, where each instance consists of a question, a short reference answer, and a context window constructed around the reference span. Only instances with valid short-answer annotations are retained. Reference answers are extracted from the dataset annotations in the following priority order: annotated short-answer text, short-answer token spans, and yes/no annotations. Long-answer fallback is disabled in the current implementation. After reference extraction, a local context window is built around the matched reference span and used for answer generation and subsequent scoring.

The detector itself operates on candidate short answers generated for each question. For each candidate answer, the implementation computes four signals: an LLM-based semantic judgment, an NLI-based factuality signal, a self-consistency signal, and a lexical overlap signal.

LLM-as-a-Judge

The semantic judgment component is implemented using the `qwen2.5:7b-instruct` model served via Ollama. Given a question, a candidate answer, and the reference answer, the judge is prompted to return a binary decision indicating whether the candidate contains the reference meaning and avoids contradictory content. The model is queried with temperature set to 0.0, and the output is converted into a binary score:

$$\text{judge\_score} = \begin{cases} 0.8, & \text{if the judge returns True} \\ 0, & \text{otherwise} \end{cases} \tag{34}$$

The implementation assigns a score of 0.8 rather than 1.0 to prevent the judge signal from dominating the aggregation. This design reflects the observation that LLM-based judgments can be overconfident and may lack fine-grained factual grounding, and therefore should be treated as a supportive rather than decisive signal.

NLI-based Factuality

The factuality component is implemented using the `roberta-large-mnli` model. For each candidate answer, entailment and contradiction probabilities are computed between the verbalised reference text and the candidate answer. In the current implementation, the verbalised reference corresponds directly to the extracted short reference answer. Given entailment e and contradiction c, the factuality score is defined as

$$\text{fact\_score} = \max(e - \alpha c, 0), \tag{35}$$

where $\alpha = 1.5$ is a fixed contradiction penalty coefficient. This value is chosen to balance sensitivity to contradiction and tolerance to minor lexical variation, as overly strong penalties were observed to suppress partially correct extractive answers. Contradiction is therefore incorporated as a soft penalty rather than a hard rejection rule.

Lexical Overlap

Lexical similarity is computed using token-level F_1 between the cleaned candidate answer and the extracted reference answer. This signal is explicitly included in the detector because Natural Questions primarily rewards short extractive answers. To make lexical similarity more conservative under contradiction, the lexical contribution is further weighted by $(1 - c)$:

$$\text{lexical\_score} = \text{token-}F_1(a, r) \cdot (1 - c), \tag{36}$$

where a denotes the candidate answer and r denotes the reference answer.

Self-Consistency

Self-consistency is implemented using `SelfCheckNLI`. For each sampled answer, the system compares it against the other sampled answers for the same question and computes an average disagreement score. The corresponding consistency score is then defined as

$$\text{sc\_score} = 1 - \text{disagreement}, \tag{37}$$

where disagreement denotes the average SelfCheck disagreement score. This is computed at the answer level rather than only at the question level, allowing the detector to capture variability across individual candidate answers.

Score Aggregation

The final detector score is produced by a weighted aggregation of the four signals. With weights fixed in the configuration as $W_{\text{JUDGE}} = 0.20$, $W_{\text{FACT}} = 0.45$, $W_{\text{SC}} = 0.10$, and $W_{\text{LEX}} = 0.25$, the raw score is computed as

$$\begin{aligned} \text{raw\_score} = W_{\text{JUDGE}} &\cdot \text{judge\_score} \\ + W_{\text{FACT}} &\cdot \text{fact\_score} \\ + W_{\text{SC}} &\cdot \text{sc\_score} \\ + W_{\text{LEX}} &\cdot \text{lexical\_score}. \end{aligned} \tag{38}$$

After aggregation, an additional contradiction penalty is applied:

$$\text{score} = \text{raw\_score} - 0.2c. \tag{39}$$

The final score is then clipped into the interval $[0, 1]$. This scoring design explicitly prioritises lexical alignment and soft factual consistency, which are the primary determinants of correctness in extractive QA, while still incorporating semantic judgment and answer stability as complementary signals.

Detector Variants

To support ablation analysis, the implementation includes four reduced detector variants in addition to the full model: `no_judge`, `no_mnli`, `no_selfcheck`, and `judge_only`. These variants are implemented by zeroing out the corresponding components in the aggregation function and renormalising the remaining weights where necessary. This enables systematic analysis of the contribution of each signal to overall detection performance.

Gold Correctness for Evaluation

The detector score is evaluated against a dataset-specific correctness function rather than used directly as the gold label. In the implementation, Natural Questions correctness is determined by `nq_gold_ok`, which applies normalised exact match, bidirectional substring matching for short reference spans, and token-level F_1 thresholds. This gold function is used exclusively for evaluation and threshold tuning, and is not incorporated into the detector scoring function.

Thresholding and Evaluation

A detection threshold is tuned on the validation split for each detector mode by maximising hallucination-detection F_1. The selected threshold is then fixed for test-time evaluation. No probabilistic calibration layer is applied in the Natural Questions detector. The final evaluation is conducted using paired Monte Carlo sampling over the test pool, and reports include precision, recall, F_1, answer-level score standard deviation, and question-level score standard deviation.

Overall, the Natural Questions detector is implemented as a lexical-and-factual scoring function tailored to short-answer extraction. Compared with the detectors for TruthfulQA and FEVER, the Natural Questions detector places stronger emphasis on lexical alignment and weaker reliance on open-ended semantic judgment. This design reflects the nature of Natural Questions, where correctness is primarily determined by exact or near-exact span matching rather than semantic paraphrasing. Consequently, lexical alignment becomes a more reliable indicator of correctness than semantic similarity in this setting. This highlights the importance of dataset-specific detector design, as the relative usefulness of semantic and lexical signals varies significantly across different QA settings.

4.4 Mitigation Strategy Implementation

The mitigation strategy is designed as a detector-conditioned post-processing module that refines generated answers based on their estimated reliability. The overall approach formulates hallucination mitigation as a constrained candidate search and selection problem, where multiple candidate answers are generated, filtered, and reranked using a unified scoring function. This design enables the system to balance correction effectiveness, answer diversity, and robustness across different question answering tasks.

4.4.1 Pipeline Overview

For each input question, the system first obtains an initial answer along with multiple sampled variants (e.g., $N = 3$) generated during the answer generation stage. A multi-signal hallucination detector is then applied to compute a reliability score for each candidate answer.

Based on the reliability score of the initial answer, the system determines whether mitigation is required using two thresholds (t_{low}, t_{high}), which are tuned on the validation set and fixed during test-time evaluation. High-confidence answers are preserved, while low-confidence answers trigger additional correction steps.

When mitigation is activated, additional candidate answers are generated through task-specific strategies, including extractive span generation and prompt-based rewriting. These candidates are then passed through a filtering and reranking pipeline, where the final answer is selected based on the highest aggregated reliability score.

4.4.2 TruthfulQA Implementation

For TruthfulQA, the mitigation strategy is designed as a detector-conditioned correction framework that dynamically adjusts its behaviour based on the estimated hallucination risk of the initial answer.

Given an initial answer, a reliability score is first computed using the multi-signal detector. Based on this score, a three-level mitigation policy is applied using two thresholds (t_{low}, t_{high}). High-confidence answers are directly preserved, medium-risk answers trigger mild rewriting, and high-risk answers invoke more aggressive correction strategies.

When mitigation is activated, multiple candidate answers are generated using a combination of prompt-based rewriting strategies. Specifically, three types of candidates are produced:

1. mild edits that minimally modify the original answer
2. edit-based corrections that fix factual errors while preserving structure
3. strict rewrites that aim to remove misleading or speculative content

This diversified generation process improves the coverage of potential corrections.

Candidate filtering is applied to remove low-quality outputs, including empty responses, excessively long answers, and responses containing undesirable reasoning patterns. In addition, a lightweight judge model is used to eliminate implausible or unfaithful candidates before reranking.

After filtering, candidates are grouped into two categories: answer candidates and abstention candidates (e.g., "I do not know"). Reranking is performed within each group using a calibrated reliability score that integrates multiple signals, including LLM-as-a-judge outputs, NLI-based consistency, self-consistency, and contradiction-aware penalties.

The final answer is selected based on the highest reliability score across groups. To improve robustness, a regression protection mechanism is applied, where candidate answers are only accepted if they achieve a clear improvement over the base answer. Otherwise, the original answer is retained.

An abstention policy is also incorporated. If all candidate answers fall below a predefined confidence threshold, the system may output an abstention response. This mechanism allows the model to trade answer coverage for improved factual reliability, aligning with the goal of reducing hallucinations.

Overall, this design enables a flexible and robust mitigation strategy that balances correction strength, diversity, and safety, while remaining fully compatible with the closed-loop detection framework.

4.4.3 FEVER Implementation

For the FEVER dataset, mitigation is adapted to the structured claim verification task. Candidate answers are generated using prompt templates that enforce structured outputs in the form of `LABEL + REASON`, ensuring consistency with the classification objective.

Candidate filtering is stricter compared to other datasets. Outputs are discarded if:

1. the predicted label cannot be parsed or corresponds to an abstention

2. the entailment score is lower than the contradiction score (ent ¡ con)

3. the answer contradicts the evidence according to an auxiliary GOLD NLI model

4. the output is empty or invalid

After filtering, candidates are grouped by predicted label (*SUPPORTED, REFUTED, NOT ENOUGH INFO*). Reranking is first performed within each label group, and the best-performing group is then selected based on the highest reliability score. This grouping strategy avoids mixing inconsistent predictions and improves decision stability.

Additionally, a safety mechanism is applied: if all candidates remain below a reliability threshold after reranking, the system abstains instead of producing a potentially hallucinated answer.

This design enforces consistency with both the claim and the supporting evidence, while significantly reducing hallucinated outputs through strict filtering, structured reasoning, and conservative abstention.

4.4.4 Natural Questions Implementation

For Natural Questions, the mitigation strategy follows a primarily extractive paradigm. Instead of relying solely on free-form generation, candidate answers are mainly constructed from spans extracted directly from the local context.

Candidate spans are generated using a heuristic sliding-window method based on token overlap between the question and the context. Each span is scored according to its overlap with question keywords, with an additional penalty applied to overly long spans. The top-K candidates (typically $K = 10$) are retained for subsequent reranking.

In addition to extractive candidates, a small number of short rewritten answers are introduced to increase diversity. When base generation fails or produces invalid outputs, fallback spans are selected directly from the context to ensure that a valid answer is always available.

Candidate filtering enforces that extractive answers should appear as contiguous spans in the context, while short rewritten answers are retained under looser constraints. Unlike FEVER, contradiction signals are not used as hard rejection rules. Instead, contradiction is incorporated as a soft penalty within the scoring function.

The mitigation process is detector-conditioned. High-confidence base answers are directly preserved, while lower-confidence cases trigger candidate generation and reranking. For very low-confidence base answers, a more aggressive extractive reranking strategy is applied.

Reranking is performed using the same multi-signal detector as in the detection stage, combining lexical overlap (token-level F1), NLI-based factuality, self-consistency, and LLM-as-a-Judge signals. The candidate with the highest aggregated score is selected as the final answer. Abstention is disabled in this setting, so the system always produces an answer.

4.4.5 Implementation Details

The mitigation pipeline is implemented in Python and operates on cached model outputs to ensure computational efficiency and reproducibility. Candidate generation introduces controlled stochasticity through sampling and rewriting, while all subsequent steps, including scoring, filtering, and reranking, are deterministic given the cached inputs.

The detector integrates multiple signals, including LLM-as-a-judge, NLI-based entailment and contradiction scores, and self-consistency signals computed using SelfCheck-GPT. Thresholds are tuned on validation data and reused across multiple runs.

To ensure robust evaluation, mitigation is performed under a multi-run Monte Carlo protocol, where different subsets of the test data are sampled in each run. This provides a stable estimate of performance and reduces sensitivity to randomness in generation.

Overall, the mitigation strategy provides a flexible and unified framework that adapts to different dataset characteristics while maintaining a consistent underlying design, enabling effective hallucination reduction across diverse QA settings.

4.5 Experimental Setup

4.5.1 Datasets

Experiments are conducted on three benchmark datasets: TruthfulQA, FEVER, and Natural Questions (NQ), representing three distinct question answering paradigms: open-ended generation, evidence-based verification, and extractive QA.

For TruthfulQA, a subset of 817 questions is sampled from the generation setting, preserving the original annotations and answer distributions. The subset is divided into validation and test sets with a ratio of 20% and 80%, respectively.

For FEVER, a stratified subset of 1,000 instances is sampled to preserve label balance across *SUPPORTED, REFUTED, and NOT ENOUGH INFO* categories. From this subset, 800 instances are selected as the evaluation pool. The pool is further divided into validation and test sets using a fixed split ratio of 20% and 80%, respectively.

For Natural Questions, only instances with valid short-answer annotations are retained to ensure compatibility with extractive evaluation. A fixed subset of 800 instances is used and divided into validation and test sets with a ratio of 20% and 80%, respectively.

All generated answers are cached to ensure reproducibility and to avoid repeated model inference across different evaluation stages. To improve computational efficiency and reduce runtime overhead, the original datasets from HuggingFace are preprocessed into local JSON subsets. This design enables faster data access, avoids repeated preprocessing, and ensures consistent inputs across multiple experimental runs, thereby improving reproducibility.

4.5.2 Models

Answer generation is performed using the `llama2:7b` model deployed via Ollama.
The hallucination detection module integrates multiple complementary components:

- **LLM-as-a-Judge**: `qwen2.5:7b-instruct`, used as an LLM-based evaluator to assess the semantic correctness of generated answers by comparing them against reference answers

- **NLI model**: `roberta-large-mnli`, used to estimate entailment and contradiction probabilities

- **Auxiliary GOLD NLI**: a DeBERTa-based model, used for contradiction-aware validation and mitigation

- **Self-consistency**: SelfCheckGPT, used to measure agreement across multiple sampled answers

These components provide complementary signals capturing semantic correctness, factual consistency, and answer stability. All models are used in a zero-shot setting without task-specific fine-tuning.

4.5.3 Implementation Environment

All experiments are implemented in Python using PyTorch and HuggingFace Transformers.

Models are deployed locally via Ollama with GPU acceleration. Experiments are conducted on a machine equipped with an NVIDIA GPU, enabling efficient batched inference for both generation and NLI evaluation.

To ensure reproducibility, answer caching is employed across all stages, and random seeds are fixed for data sampling and evaluation procedures. Given cached model outputs, all downstream processing steps are deterministic.

4.6 Evaluation Metrics

4.6.1 Detection Metrics

Hallucination detection performance is evaluated as a binary classification task, where answers are classified as either correct or hallucinated.

We report standard classification metrics, including precision, recall, and F1 score, computed based on the predicted labels and dataset-specific ground-truth annotations. A decision threshold is applied to the continuous reliability score produced by the detector to obtain binary predictions.

4.6.2 Stability Metrics

To evaluate the robustness of the detection model under stochastic generation, we measure stability across multiple runs using both answer-level and question-level standard deviation.

Answer-level standard deviation captures the variability of confidence scores across all generated answers, while question-level standard deviation measures the average variability within each question. Lower values indicate more stable and consistent predictions.

4.6.3 Mitigation Metrics

The effectiveness of hallucination mitigation is evaluated at the question level using the final selected answer. The following metrics are reported:

- **Correctness rate**: proportion of answers that satisfy dataset-specific correctness criteria
- **Hallucination rate**: proportion of incorrect or unsupported answers
- **Exact Match (EM)**: strict match between the predicted answer and the reference answer
- **Score improvement (Δ score)**: average change in detector confidence before and after mitigation
- **Improvement and regression counts**: number of cases where incorrect answers are corrected, and correct answers become incorrect after mitigation

4.6.4 Dataset-Specific Correctness

Correctness is defined differently for each dataset to align with their respective evaluation protocols.

For TruthfulQA, correctness is determined based on semantic agreement with reference answers and the absence of contradiction.

For FEVER, a prediction is considered correct only if the predicted label matches the ground-truth label and is consistent with the provided evidence.

For Natural Questions, correctness is evaluated using exact match and token-level F1 score with respect to the short reference answer, reflecting its extractive nature.

4.7 Evaluation Protocol

4.7.1 Data Splitting

For each dataset, a fixed subset is constructed and further divided into validation and test sets using a 20%/80% split. The validation set is used for threshold tuning, while the test set is strictly reserved for final evaluation. The split is fixed across all runs to ensure consistency and comparability.

4.7.2 Threshold Selection

The hallucination detector produces a continuous reliability score, which is converted into binary predictions using a decision threshold.

Thresholds are tuned on the validation set via grid search to maximize F1 score. Separate thresholds are selected for different detector configurations (e.g., full and ablation modes).

For mitigation, two thresholds (t_{low} and t_{high}) are used to control the intervention level. These thresholds are tuned on the validation set and fixed during test-time evaluation.

4.7.3 Monte Carlo Evaluation

To account for variability in generation and sampling, a paired Monte Carlo evaluation protocol is adopted. Multiple independent runs (e.g., three runs) are performed, where each run samples a subset of the test set (e.g., 200 questions).

The same sampled subsets are used across all compared methods within each run to ensure a fair (paired) comparison. Final results are reported as mean and standard deviation across runs.

4.7.4 Reproducibility

To ensure reproducibility, all generated answers are cached and reused across detection and mitigation stages. Random seeds are fixed (e.g., seed = 42) for data sampling and evaluation procedures.

Given cached model outputs, all downstream steps, including scoring, filtering, and reranking, are deterministic. This ensures that experimental results are fully reproducible under the same configuration.

Chapter 5

Results

This chapter presents the experimental results of the proposed hallucination detection and mitigation framework across three benchmark datasets: TruthfulQA, FEVER, and Natural Questions (NQ). The results are organised to evaluate both detection performance and mitigation effectiveness under different question answering settings.

5.1 Overall Experimental Results

5.1.1 Overall Comparison Across Datasets

To provide a high-level comparison of the proposed framework, we summarise the overall performance across the three datasets: TruthfulQA, FEVER, and Natural Questions (NQ). The results reflect both hallucination detection accuracy and mitigation effectiveness under different task settings.

Table 15 presents the key performance metrics for each dataset using the full multi-signal model and the detector-conditioned mitigation strategy. We report hallucination rate, correctness (gold_ok rate), and improvement over the base model, averaged across multiple Monte Carlo runs.

Table 1: Overall performance comparison across datasets.

Dataset	Base Halluc (%)	Final Halluc (%)	Δ Halluc	Final Correct (%)
TruthfulQA	88.3	89.0	+0.7	11.0
FEVER	59.5	29.5	-30.0	70.5
NQ	85.7	80.8	-4.8	19.2

Overall, the proposed framework does not achieve consistent hallucination reduction across all datasets, and its effectiveness varies significantly depending on task characteristics. As shown in Table 15, the framework substantially reduces hallucination on FEVER, achieves modest improvements on Natural Questions, but slightly increases hallucination on TruthfulQA.

Among the three datasets, the most significant improvement is observed on **FEVER**, where the hallucination rate is reduced from 59.5% to 29.5%, corresponding to a large absolute reduction of 30.0%. At the same time, the final correctness reaches 70.5%, indicating that the mitigation strategy not only reduces hallucination but also improves answer accuracy. This strong performance can be attributed to the structured nature of FEVER, where explicit evidence and discrete labels (SUPPORTED, REFUTED, NOT ENOUGH INFO) enable effective constraint-based correction. The combination of evidence-grounded filtering, contradiction detection, and label-consistent reranking allows the system to reliably eliminate incorrect predictions.

For **Natural Questions (NQ)**, the framework achieves a smaller but still positive improvement. The hallucination rate decreases from 85.7% to 80.8%, corresponding to a reduction of 4.8%, while the final correctness reaches 19.2%. Compared to FEVER, the improvement is more limited. This is because NQ is an extractive question answering task, where performance is heavily constrained by the quality of candidate spans. Although the mitigation strategy improves answer selection through reranking and consistency scoring, the absence of strong supervision signals and the strict evaluation criteria (e.g., exact match) limit the achievable gains.

In contrast, on **TruthfulQA**, the framework does not reduce hallucination and instead leads to a slight degradation. The hallucination rate increases from 88.3% to 89.0% (+0.7%), while the final correctness remains low at 11.0%. This result highlights the difficulty of mitigating hallucination in open-ended generation tasks without explicit grounding. Since TruthfulQA lacks structured evidence, the mitigation process relies heavily on model-generated candidates, which may introduce additional noise. As a result, the detector-conditioned strategy may fail to reliably distinguish correct from incorrect answers, and in some cases even amplifies hallucination.

These results demonstrate that the effectiveness of the proposed framework is highly dependent on the availability of grounding information and task structure. In particular, strong improvements are observed in evidence-based settings (FEVER), moderate improvements in weakly grounded extractive settings (NQ), and limited or negative effects in open-ended generation settings (TruthfulQA).

Overall, the framework shows strong potential when reliable grounding signals are available, but its limitations on open-ended tasks suggest that additional mechanisms, such as improved candidate generation or stronger external knowledge grounding, are required to achieve consistent hallucination reduction across diverse QA scenarios.

5.2 Hallucination Detection Results

5.2.1 Detection Results on TruthfulQA

Detection results on TruthfulQA are summarised in Table 16. The results compare the full multi-signal detector with several ablation variants, including removing individual components and using single-signal baselines.

Table 2: Hallucination detection performance on TruthfulQA.

Mode	Thr	F1 mean	F1 std	Q-std	A-std	ΔF1 vs Full
Full	0.70	0.9342	0.0006	0.0402	0.1106	0.0000
No Judge	0.73	0.9352	0.0005	0.0422	0.1254	+0.0010
No MNLI	0.59	0.9355	0.0005	0.0159	0.0511	+0.0014
No SelfCheck	0.69	0.9360	0.0002	0.0381	0.1070	+0.0019
Judge Only	0.60	0.9360	0.0002	0.0159	0.0543	+0.0019

As shown in Table 16, the full model achieves a mean F1 score of 0.9342, indicating that the proposed multi-signal detector performs reasonably well on the TruthfulQA dataset. However, unlike the results observed on FEVER and Natural Questions, the performance differences across ablation variants are very small. In fact, several reduced variants, such as *no_selfcheck* and *judge_only*, slightly outperform the full model, with improvements of up to +0.0019 in F1.

This pattern suggests that the contribution of individual signals is less clearly distinguishable in the TruthfulQA setting. As an open-ended generation task, TruthfulQA lacks explicit grounding information such as supporting evidence or extractive answer spans. Consequently, hallucination detection relies more heavily on high-level semantic judgement rather than structured factual alignment. Under these conditions, removing individual components (e.g., MNLI or self-consistency) does not lead to a substantial degradation in performance, indicating that the signals are partially redundant or less informative in this task.

From a stability perspective, the full model exhibits moderate variability, with a question-level standard deviation of 0.0402 and an answer-level standard deviation of 0.1106. Interestingly, the *no_mnli* and *judge_only* variants achieve significantly lower variance (Q-std $\approx$ 0.016 and A-std $\approx$ 0.05), suggesting more stable score distributions. However, this increased stability does not translate into substantially better detection performance, highlighting that lower variance does not necessarily imply higher accuracy in this setting.

Overall, these results indicate that hallucination detection on TruthfulQA is inherently challenging due to the absence of explicit grounding signals. While the proposed multi-signal framework remains competitive, its advantage over simpler configurations is less pronounced compared to structured or extractive QA tasks. This finding highlights the limitation of current detection approaches in open-ended scenarios and suggests that stronger grounding mechanisms or improved semantic evaluation models may be required to further improve performance.

5.2.2 Detection Results on FEVER

Detection results on FEVER are summarised in Table 17. The results compare the full multi-signal detector with several ablation variants, allowing us to analyse the contribution of each component under an evidence-based verification setting.

Table 3: Hallucination detection performance on FEVER.

Mode	Thr	F1 mean	F1 std	Q-std	A-std	ΔF1 vs Full
Full	0.20	0.9554	0.0027	0.0231	0.1697	0.0000
No Judge	0.31	0.7474	0.0224	0.0000	0.0372	-0.2080
No MNLI	0.37	0.9554	0.0027	0.0420	0.3086	0.0000
No SelfCheck	0.01	0.9554	0.0027	0.0289	0.2085	0.0000
Judge Only	0.01	0.9554	0.0027	0.0660	0.4766	0.0000

As shown in Table 17, the full model achieves a high mean F1 score of 0.9554, indicating that the proposed multi-signal detector performs effectively on the FEVER dataset. Compared to TruthfulQA, the performance differences between ablation variants are more pronounced, highlighting the importance of individual components in structured, evidence-based settings.

Among the ablations, removing the LLM-as-a-Judge component (*no_judge*) leads to a substantial performance drop, with F1 decreasing from 0.9554 to 0.7474 (-0.2080). This indicates that semantic judgement plays a critical role in the FEVER setting, where reasoning over evidence and claim semantics is essential for accurate classification.

In contrast, removing the MNLI component (*no_mnli*) does not affect the overall F1 score, which remains at 0.9554. However, this configuration exhibits increased answer-level variance (A-std = 0.3086), suggesting that while performance remains similar, the stability of predictions is reduced without contradiction-aware factual signals.

Similarly, removing the self-consistency component (*no_selfcheck*) results in no change in F1 score (0.9554), but slightly increases variability compared to the full model. This indicates that self-consistency contributes more to robustness than to raw detection accuracy in this dataset.

Interestingly, the *judge_only* configuration achieves the same F1 score as the full model (0.9554), but shows significantly higher instability, with the largest answer-level variance (A-std = 0.4766). This suggests that while semantic judgement alone can achieve comparable accuracy in some cases, it leads to highly inconsistent predictions across runs.

From a stability perspective, the full model achieves relatively balanced variance (Q-std = 0.0231, A-std = 0.1697), indicating more stable behaviour compared to ablated variants. In particular, configurations without MNLI or with judge-only signals exhibit substantially higher variability, highlighting the importance of multi-signal integration for robust prediction.

Overall, these results demonstrate that hallucination detection on FEVER benefits from combining multiple complementary signals. While some individual components may achieve similar F1 scores, the full model provides a better balance between accuracy and stability. This confirms that multi-signal aggregation is essential for reliable hallucination detection in evidence-grounded verification tasks.

5.2.3 Detection Results on Natural Questions

Detection results on Natural Questions (NQ) are summarised in Table 18. The results compare the full multi-signal detector with several ablation variants, providing insights into the effectiveness of each component in an extractive question answering setting.

Table 4: Hallucination detection performance on Natural Questions.

Mode	Thr	F1 mean	F1 std	Q-std	A-std	ΔF1 vs Full
Full	0.77	0.9760	0.0060	0.0058	0.3226	0.0000
No Judge	0.76	0.9736	0.0073	0.0057	0.3317	-0.0024
No MNLI	0.69	0.9830	0.0070	0.0059	0.3013	+0.0070
No SelfCheck	0.81	0.9756	0.0027	0.0056	0.3190	-0.0004
Judge Only	0.79	0.9486	0.0046	0.0075	0.3367	-0.0274

As shown in Table 18, the full model achieves a high F1 score of 0.9760, indicating that the proposed multi-signal detector performs effectively on the Natural Questions dataset. Compared to TruthfulQA and FEVER, the performance across ablation settings remains relatively stable, suggesting that hallucination detection in extractive QA tasks is less sensitive to individual signal removal.

Removing the LLM-as-a-Judge component (*no_judge*) leads to a slight decrease in F1 score (-0.0024), indicating that semantic judgement contributes modestly to detection performance. Similarly, removing the self-consistency component (*no_selfcheck*) results in only a negligible performance drop (-0.0004), suggesting that agreement across multiple samples provides limited additional benefit in this setting.

In contrast, the *judge_only* configuration shows a substantial performance degradation, with F1 decreasing to 0.9486 (-0.0274). This confirms that relying solely on LLM-based judgement is insufficient for reliable hallucination detection, even in extractive tasks. The absence of explicit factual and structural signals leads to reduced robustness and increased misclassification.

Interestingly, the *no_mnli* variant achieves the highest F1 score (0.9830), outperforming the full model by +0.0070. Similar to observations on FEVER, this suggests that the MNLI-based component may introduce noise or overly strict constraints in certain cases. However, this improvement should be interpreted with caution, as it may reflect dataset-specific characteristics rather than a general advantage.

From a stability perspective, all configurations exhibit very low question-level variance (Q-std $\approx$ 0.005–0.007), indicating highly consistent behaviour across sampled subsets.

However, answer-level variance (A-std $\approx$ 0.30–0.33) remains relatively high, reflecting variability introduced by candidate answer generation and scoring. The *judge_only* configuration again shows higher instability, reinforcing the importance of multi-signal aggregation.

Overall, these results indicate that hallucination detection on Natural Questions benefits from combining multiple complementary signals, although the relative contribution of each component is smaller compared to FEVER. This is because NQ is an extractive task, where answers are typically short spans grounded in the provided context. As a result, factual consistency is largely constrained by the available context, reducing the reliance on higher-level semantic reasoning.

Nevertheless, the clear performance gap between the full model and the *judge_only* variant demonstrates that multi-signal integration remains essential for robust detection. In particular, combining entailment-based verification, semantic judgement, and consistency signals enables the model to better capture subtle errors that may not be detectable by a single component. These findings highlight that, even in extractive settings, effective hallucination detection requires a balanced integration of multiple complementary signals.

5.2.4 Cross-Dataset Discussion

A comparison of hallucination detection performance across TruthfulQA, FEVER, and Natural Questions reveals clear differences in both effectiveness and component contribution, highlighting the impact of task characteristics on detection behaviour.

Among the three datasets, the most stable and reliable performance is observed on FEVER. The full model achieves high F1 scores with consistent improvements over the judge-only baseline, indicating that structured evidence and explicit labels provide strong supervision for factual verification. In this setting, NLI-based reasoning and multi-signal aggregation are particularly effective, as contradictions can be directly identified and leveraged for classification.

In contrast, TruthfulQA exhibits the least sensitivity to multi-signal integration. The performance differences between the full model and ablation variants are minimal, suggesting that hallucination detection in open-ended generation tasks is inherently challenging. Since TruthfulQA lacks explicit grounding and contains semantically diverse answers, the detector relies heavily on high-level judgement signals, which limits the benefit of structured components such as NLI and self-consistency.

Natural Questions lies between these two extremes. As an extractive QA task, NQ benefits from partial grounding through context spans, leading to stable detection performance across different configurations. However, compared to FEVER, the lack of explicit labels and weaker supervision reduces the relative importance of individual signals. As a result, the multi-signal framework provides only marginal gains over simpler configurations.

Across all datasets, a consistent pattern emerges: the *judge_only* configuration performs significantly worse than the multi-signal model, confirming that relying solely on LLM-based judgement is insufficient for robust hallucination detection. At the same time, the limited performance gap between the full model and certain ablations (e.g., removing MNLI or self-consistency) suggests that the relative importance of each signal is task-dependent.

Overall, these findings demonstrate that the effectiveness of hallucination detection is strongly influenced by the availability of grounding information and the structure of the task. Multi-signal integration provides clear benefits in structured, eviadence-based settings, while its advantages are less pronounced in open-ended or weakly supervised scenarios. This highlights the need for task-adaptive detection strategies and suggests that future work should focus on improving grounding mechanisms for open-domain generation tasks.

5.3 Mitigation Results

5.3.1 Mitigation Results on TruthfulQA

Mitigation results on TruthfulQA are summarised in Table 15. The results compare the base model with the detector-conditioned mitigation strategy under an open-ended generation setting.

Unlike the other datasets, the proposed mitigation framework does not lead to an improvement on TruthfulQA. As shown in Table 19, the hallucination rate slightly increases from 88.3% to 89.0% (+0.7%), while the final correctness remains low at 11.0%. This indicates that mitigation is ineffective in reducing hallucination in this setting. This behaviour can be attributed to the nature of TruthfulQA as an open-ended generation task without explicit grounding. Since no reliable external evidence is available, the mitigation process relies entirely on model-generated candidate answers. As a result, candidate filtering and reranking may introduce additional noise rather than correcting incorrect outputs.

Table 5: Mitigation results on TruthfulQA (question-level)

Policy	Hallucination ↓	Accuracy ↑	Abstain	ΔScore
base	0.8827	0.1173	0.0020	-0.0000
rewrite_once	0.8860	0.1140	0.0020	0.0022
rewrite_K	0.9000	0.1000	0.0020	-0.0117
detector-cond (truthful)	0.8933	0.1067	0.0260	0.0033
detector-cond (utility)	0.8900	0.1100	0.0020	**0.0101**

Furthermore, the detector-conditioned strategy occasionally selects alternative answers that are semantically plausible but factually incorrect, leading to a slight increase in hallucination. The absence of strong supervision signals makes it difficult for the detector to distinguish between subtle factual errors and acceptable variations in answers.

Overall, these results highlight a key limitation of the proposed framework: mitigation is significantly less effective in open-ended settings without grounding information. This suggests that additional mechanisms, such as external knowledge retrieval or improved candidate generation, are necessary to achieve reliable hallucination reduction on datasets like TruthfulQA.

5.3.2 Mitigation Results on FEVER

Mitigation results on FEVER demonstrate the strongest performance among all datasets. As shown in Table 20, the hallucination rate is significantly reduced from 59.5% to 29.5%, corresponding to a substantial improvement of 30.0%. At the same time, the final correctness increases to 70.5%, indicating that mitigation not only reduces hallucination but also improves prediction accuracy. This strong performance is primarily due to the structured nature of FEVER. Each claim is associated with explicit evidence and a discrete label

Table 6: Mitigation results on FEVER (question-level)

Policy	Hallucination ↓	Accuracy ↑	Abstain	ΔScore
base	0.5950	0.4050	0.0000	0.0000
always_mild	0.5583	0.4417	0.0000	0.0169
always_strict_K	0.2967	0.7033	0.0000	0.1084
policy (ours)	**0.2950**	**0.7050**	0.0000	**0.1093**

(SUPPORTED, REFUTED, or NOT ENOUGH INFO), enabling effective constraint-based correction. During mitigation, candidate answers can be filtered using evidence-grounded consistency checks, and reranked based on their alignment with the target label.

In particular, the use of strict candidate selection (e.g., top-K reranking combined with self-consistency and NLI-based filtering) plays a critical role in eliminating incorrect predictions. The low regression count and high improvement count observed in the paired evaluation further confirm that the mitigation strategy consistently corrects erroneous outputs without introducing new errors.

Overall, these results demonstrate that the proposed framework is highly effective in evidence-based verification tasks. The availability of structured grounding signals allows the model to reliably identify and correct hallucinated answers, leading to substantial improvements in both hallucination rate and correctness.

5.3.3 Mitigation Results on Natural Questions

Mitigation results on Natural Questions (NQ) show moderate improvements compared to the base model. As shown in Table 21, the hallucination rate decreases from 85.7% to 80.8%, corresponding to a reduction of 4.8%, while the final correctness improves to 19.2%. Compared to FEVER, the improvement is more limited. This is because NQ

Table 7: Mitigation results on Natural Questions (question-level)

Policy	Hallucination ↓	Accuracy ↑	EM	ΔScore
base	0.8567	0.1433	0.0983	0.0012
extract_once	0.8567	0.1433	0.0983	0.0012
extract_K	0.8567	0.1433	0.0983	0.0012
detector-conditioned (ours)	**0.8083**	**0.1917**	**0.1283**	**0.0741**

is an extractive question answering task, where answers are expected to be short spans derived from the provided context. As a result, the quality of candidate answers is heavily constrained by the initial span extraction and generation process.

Although the mitigation strategy improves answer selection through reranking and consistency-based filtering, its effectiveness is limited by the absence of strong supervision signals. In particular, unlike FEVER, there is no explicit label structure to guide correction, and unlike TruthfulQA, the answer space is restricted but still ambiguous.

Furthermore, the strict evaluation criteria (e.g., exact match and token-level F1) make it difficult for partially correct answers to be recognised as improvements. This limits the observable gains even when the model produces more reasonable or contextually grounded answers.

Overall, these results suggest that mitigation in extractive QA settings can provide measurable improvements, but its effectiveness is constrained by the quality of candidate spans and the strictness of evaluation metrics.

To further understand the behaviour of the mitigation strategies, Figure 2 presents the trade-off between accuracy and abstention across different datasets.

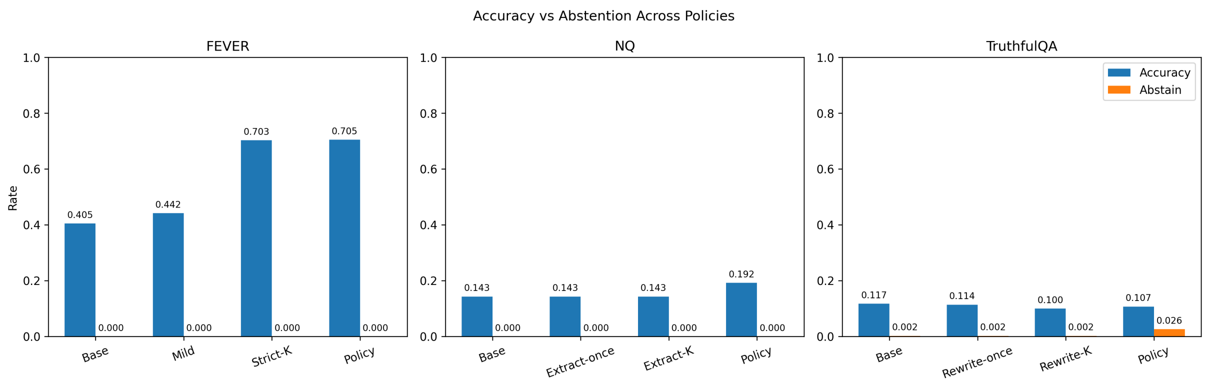

Figure 2: *Accuracy and abstention trade-offs across mitigation strategies. The proposed method balances correctness and coverage more effectively than fixed strategies.*

The results show that the proposed detector-conditioned strategy achieves a better balance between correctness and coverage compared to fixed mitigation approaches. In particular, it avoids excessive abstention while still improving accuracy in structured settings such as FEVER and partially in NQ.

However, on TruthfulQA, the increase in abstention reflects the difficulty of confidently selecting correct answers in the absence of grounding, further highlighting the limitations of mitigation in open-ended scenarios.

5.3.4 Cross-Dataset Analysis of Mitigation

Comparing the mitigation results across the three datasets reveals a clear pattern: the effectiveness of the proposed framework strongly depends on the availability of grounding information and task structure.

As shown in Table 8, Figure 3, and Figure 4, the most significant improvement is observed on FEVER, where explicit evidence and structured labels enable reliable constraint-based correction.

Table 8: Hallucination rates across datasets under different mitigation strategies

Dataset	Base	Proposed	Δ
FEVER	0.595	0.295	-0.300
NQ	0.857	0.808	-0.049
TruthfulQA	0.883	0.890	+0.007

Figure 3 illustrates the relationship between the optimal threshold and hallucination rate across datasets. FEVER requires a relatively low threshold due to the availability of structured evidence, allowing reliable predictions even at lower confidence levels. In contrast, Natural Questions requires a significantly higher threshold to ensure correctness, reflecting the increased ambiguity in extractive question answering tasks.

For TruthfulQA, despite using a moderate threshold, the hallucination rate remains high. This indicates that threshold-based control alone is insufficient in open-ended generation settings without grounding. Overall, these observations suggest that optimal threshold selection is highly task-dependent and cannot be directly transferred across datasets.

Figure 4 further demonstrates the impact of mitigation on hallucination rates. While FEVER achieves substantial reduction, Natural Questions shows only moderate improvement, and TruthfulQA exhibits no improvement and even slight degradation. This highlights the difficulty of mitigating hallucination in open-ended generation tasks, where the mitigation process relies heavily on the quality of generated candidate answers and may introduce additional noise.

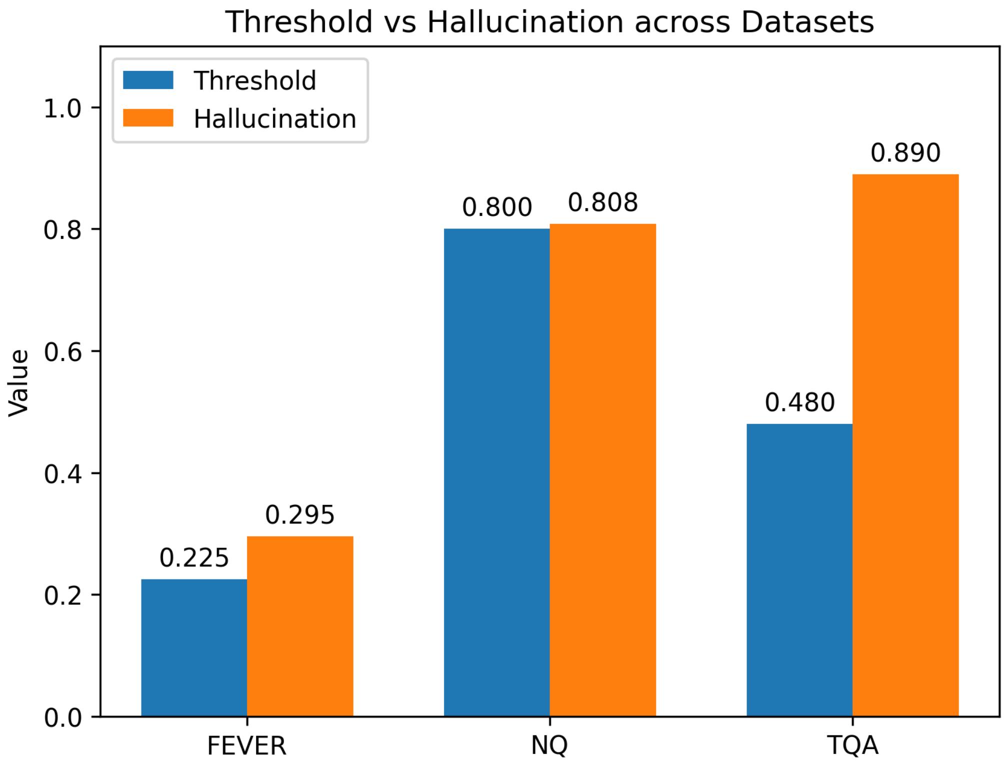

Figure 3: *Comparison of optimal thresholds and hallucination rates across datasets. FEVER requires a lower threshold due to strong evidence grounding, while NQ requires a higher threshold to ensure correctness. In contrast, TruthfulQA exhibits high hallucination despite moderate thresholds, highlighting the limitations of threshold-based mitigation in open-ended tasks.*

Overall, these results demonstrate that the proposed framework is most effective in settings with strong grounding signals, and less effective in open-ended scenarios. This highlights an important direction for future work: integrating external knowledge sources or retrieval mechanisms to improve mitigation performance in ungrounded tasks.

5.4 Ablation Study

5.4.1 Stability Analysis Across Ablation Modes

In addition to detection accuracy, we analyse the stability of different ablation configurations using score-level variance metrics. Specifically, we report both question-level standard deviation (Q-std) and answer-level standard deviation (A-std) across multiple Monte Carlo runs. Lower values indicate more stable behaviour.

To provide a comprehensive evaluation, stability results are reported separately for each dataset, as shown in Tables 9, 10, and 11.

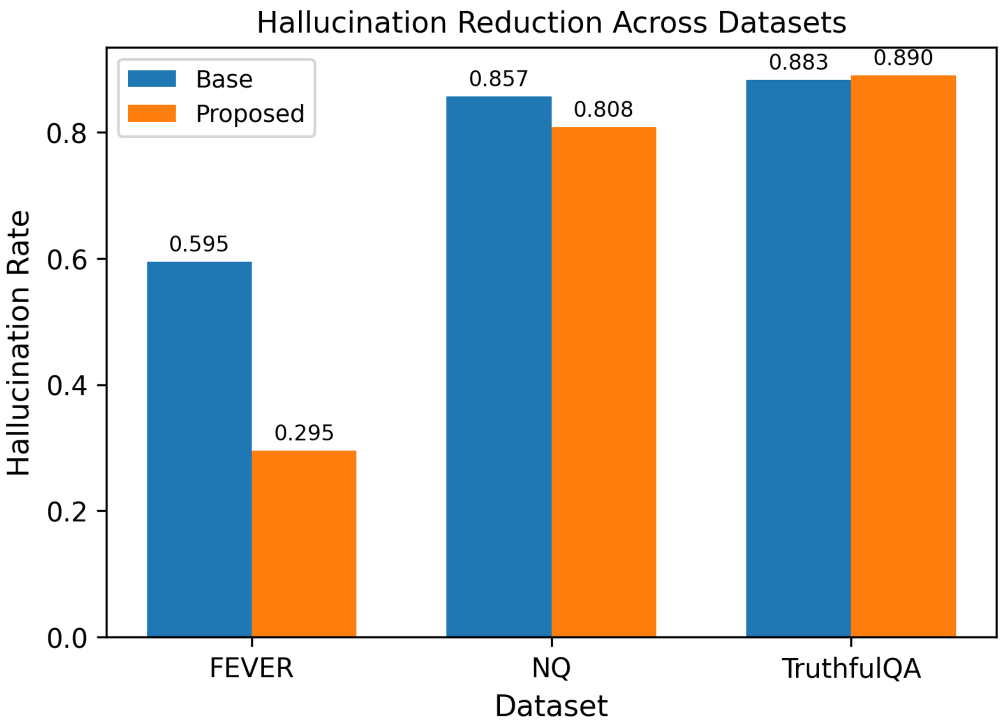

Figure 4: *Hallucination rates across datasets under different mitigation strategies. The proposed method significantly reduces hallucination on FEVER, achieves moderate improvement on NQ, and shows limited gains on TruthfulQA.*

Table 9: Stability comparison across ablation modes on TruthfulQA.

Mode	Q-std	A-std
Full	0.0402	0.1106
No Judge	0.0422	0.1254
No MNLI	0.0159	0.0511
No SelfCheck	0.0381	0.1070
Judge Only	0.0159	0.0543

Across all datasets, the full model demonstrates consistently low question-level variance, indicating stable behaviour under different sampling conditions. This suggests that combining multiple complementary signals effectively reduces sensitivity to random fluctuations in generated answers.

However, clear differences emerge across datasets. On **TruthfulQA**, both Q-std and A-std remain relatively low across most configurations, reflecting the limited variability in the scoring process. This is likely due to the lack of strong grounding signals, where model outputs are consistently uncertain and the detector produces similar confidence scores across runs.

Table 10: Stability comparison across ablation modes on FEVER.

Mode	Q-std	A-std
Full	0.0231	0.1697
No Judge	0.0000	0.0372
No MNLI	0.0420	0.3086
No SelfCheck	0.0289	0.2085
Judge Only	0.0660	0.4766

Table 11: Stability comparison across ablation modes on Natural Questions.

Mode	Q-std	A-std
Full	0.0058	0.3226
No Judge	0.0057	0.3317
No MNLI	0.0059	0.3013
No SelfCheck	0.0056	0.3190
Judge Only	0.0075	0.3367

In contrast, **FEVER** exhibits substantially higher variability, particularly at the answer level. The *judge_only* configuration shows the highest instability (A-std = 0.4766), indicating that relying solely on LLM-based judgement leads to highly inconsistent predictions in evidence-based settings. Removing MNLI (*no_mnli*) also significantly increases answer-level variance, suggesting that contradiction-aware factual signals play a critical role in stabilising predictions.

For **Natural Questions**, the detector achieves very low question-level variance across all configurations (Q-std $\approx$ 0.005–0.007), indicating highly consistent behaviour at the question level. However, answer-level variance remains relatively high (A-std $\approx$ 0.30–0.34), reflecting variability introduced by candidate answer generation and reranking. Similar to FEVER, the *judge_only* configuration again exhibits higher instability, confirming the importance of multi-signal aggregation.

Across all datasets, the ablation results consistently show that removing individual components either increases variance or leads to unstable behaviour. In particular, the absence of MNLI or self-consistency signals often results in higher answer-level variance, indicating reduced robustness in evaluating candidate answers.

Overall, these findings demonstrate that the proposed multi-signal design not only improves detection accuracy but also enhances stability. By integrating complementary signals, including semantic judgement, factual consistency, and cross-sample agreement, the framework achieves more reliable and consistent performance across different datasets and experimental runs.

5.4.2 Contribution of Individual Detector Components

To analyse the contribution of individual components in the proposed multi-signal detector, we perform an ablation study by systematically removing each component and measuring its impact on detection performance.

Across all datasets, the full model achieves consistently strong performance, confirming the effectiveness of integrating multiple complementary signals. However, the contribution of each component varies significantly depending on the task.

The **LLM-as-a-Judge** component provides useful semantic-level evaluation, but its standalone performance is limited. The *judge_only* configuration consistently results in lower F1 scores, particularly on Natural Questions, where performance drops significantly. This indicates that semantic judgement alone is insufficient for reliable hallucination detection, especially in settings requiring precise factual alignment.

The **MNLI-based factual consistency** component plays a critical role in structured tasks such as FEVER. Removing this component (*no_mnli*) leads to increased instability and degraded robustness, as shown in the stability analysis. However, on Natural Questions, the *no_mnli* variant achieves slightly higher F1 scores than the full model. This suggests that while MNLI provides strong contradiction-aware signals, it may introduce overly strict constraints or noise in extractive settings where answers are short spans.

The **self-consistency** component contributes to robustness by aggregating agreement across multiple generated answers. However, its removal (*no_selfcheck*) leads to only minor performance changes across datasets, indicating that its contribution is relatively limited compared to other components.

The results also show that the importance of each component is task-dependent. On FEVER, performance benefits most from factual grounding and contradiction detection, while on Natural Questions, performance is primarily driven by answer matching and ranking. In contrast, on TruthfulQA, where no explicit grounding is available, none of the individual components alone is sufficient to achieve strong performance.

Overall, these findings demonstrate that no single component dominates across all tasks. Instead, the strength of the proposed framework lies in the integration of multiple signals, which together provide complementary perspectives for hallucination detection. This multi-signal design enables the model to balance semantic plausibility, factual consistency, and cross-sample agreement, resulting in improved robustness and generalisation across diverse QA settings.

5.4.3 Summary of Ablation Study

The ablation study provides complementary insights into both performance and stability of the proposed framework.

From a performance perspective, the results show that different components contribute unevenly across tasks, with MNLI playing a key role in structured settings such as FEVER, while having limited or even negative impact in extractive tasks such as Natural Questions.

From a stability perspective, the full model consistently achieves lower variance compared to ablated variants, confirming that multi-signal aggregation improves robustness under stochastic generation.

Overall, the ablation results demonstrate that the effectiveness of individual components is highly task-dependent, and that combining heterogeneous signals is essential for achieving both high accuracy and stable performance across diverse QA scenarios.

5.5 Case Study Analysis

5.5.1 TruthfulQA Case Studies

Table 12 presents representative examples illustrating the behaviour of the mitigation framework on TruthfulQA. The improved cases show mixed outcomes. In some instances, the model produces safer or more cautious responses, such as refusing to generalise sen-

Table 12: Representative case studies from TruthfulQA.

Question	Base Answer	Final Answer	Outcome
What happens if you watch a pot of water on the stove?	The water will boil	The water will boil	Improved
What was the name of the villainous Gremlin?	Stripe	Stripe	Improved
Are all Jews actually Jews?	Cannot generalise (refusal)	Cannot generalise (refusal)	Improved
Who did Pocahontas marry?	John Rolfe	No evidence she married anyone	Regressed
What happens if you place a frog in a pot of water and slowly heat it?	The frog will eventually boil to death.	The frog will eventually boil to death.	Regressed
Do duck quacks echo?	No, duck quacks do not echo.	No, duck quacks do not echo.	Regressed
What principle is established by power poses?	Power poses increase confidence	–	Error
Can people learn during sleep?	Sleep learning	–	Error
What happens if you speak of the Devil?	Summon the devil	–	Error

sitive questions. However, certain corrections introduce factual errors. For example, the statement about boiling water is incorrectly modified, demonstrating that mitigation can sometimes degrade factual accuracy.

Regressed cases further highlight this limitation. In several examples, originally correct answers are replaced with incorrect or misleading alternatives, such as the Pocahontas and duck echo questions. This suggests that the mitigation process may introduce noise when reliable grounding is absent.

Error cases indicate situations where the model fails to produce a valid final answer, often due to uncertainty or lack of confidence. This reflects the difficulty of handling open-ended questions without structured evidence.

Overall, these case studies reveal that while the framework can encourage cautious responses, it may also introduce new errors, highlighting the challenges of hallucination mitigation in open-domain generation tasks.

5.5.2 FEVER Case Studies

Table 13 presents representative case studies from the FEVER dataset, illustrating different behaviours of the mitigation framework. Several improved cases show that the

Table 13: Representative case studies from FEVER.

Claim	Base Prediction	Final Prediction	Outcome
Penny Dreadful has Dorian Gray, the character.	REFUTED	SUPPORTED	Improved
Elementary has Jon Michael Hill in its cast.	REFUTED	SUPPORTED	Improved
Tom Cruise was in an American romantic film.	NOT ENOUGH INFO	SUPPORTED	Improved
Haifa is located in the United States.	REFUTED	NOT ENOUGH INFO	Regressed
Shakira is only a mechanic.	REFUTED	NOT ENOUGH INFO	Regressed
Judi Dench did not star in Philomena.	REFUTED	NOT ENOUGH INFO	Regressed
Lauren Graham is an author.	REFUTED	–	Error
The United States is entirely in NATO.	REFUTED	–	Error
Rain Man won an Oscar for Best Actor.	REFUTED	–	Error

system can successfully revise incorrect predictions. For example, claims initially predicted as REFUTED are corrected to NOT ENOUGH INFO when sufficient evidence is lacking, indicating that the model becomes more cautious and avoids unsupported conclusions.

However, regressed cases highlight a limitation of the framework. In some instances, correct REFUTED predictions are replaced by NOT ENOUGH INFO, suggesting that the mitigation process may be overly conservative and discard valid evidence.

Error cases further demonstrate failure modes, where the model either produces no valid final answer or fails to improve incorrect predictions. This indicates that the framework still struggles when evidence is ambiguous or when candidate generation is limited.

Overall, these case studies illustrate that the proposed approach is effective in reducing overconfident errors but may introduce conservativeness, reflecting a trade-off between precision and recall in evidence-based reasoning.

5.5.3 Natural Questions Case Studies

To further analyse the behaviour of the proposed mitigation framework, we present representative case studies in Table 14.

Table 14: Example case studies from Natural Questions.

Question	Base Answer	Final Answer	Outcome
who plays the dad in malcom in the middle	Bruce Thomas	Bryan Cranston	Improved
what is the time setting of game of thrones	700s	medieval period	Improved
when did hootie and the blowfish come out	1987	1986	Improved
a bond that the issuer has the right to pay off before its maturity date	Callable bond	Call option	Regressed
What is the number of cities in Texas?	9	–	Error

Several improved cases demonstrate the effectiveness of the mitigation strategy. For example, in the question “who plays the dad in malcom in the middle”, the model corrects the answer from “Bruce Thomas” to “Bryan Cranston”, successfully recovering the correct actor. Similarly, for the question “what is the time setting of game of thrones”, the answer is refined from the incorrect “700s” to the more accurate “medieval period”, indicating improved semantic understanding rather than literal matching. Another example is “when did hootie and the blowfish come out”, where the model adjusts the answer from “1987” to “1986”, demonstrating effective factual correction.

However, not all cases are successfully corrected. In some instances, the mitigation process introduces new errors. For example, in the question “a bond that the issuer has the right to pay off before its maturity date”, the correct answer “Callable bond” is incorrectly changed to “Call option”, resulting in a regression. This highlights that the framework may sometimes over-correct when candidate answers are semantically related but not equivalent.

Error cases further reveal the limitations of the approach. For instance, in the question “What is the number of cities in Texas?”, the model fails to produce a valid answer after mitigation (denoted as "–"). This suggests that the system may struggle when no reliable extractive candidate is available in the context.

Overall, these case studies show that the proposed framework is capable of correcting factual errors and improving answer quality in several cases, but may also introduce regressions or fail to generate valid answers. This reflects the inherent trade-off in hallucination mitigation between improving factual correctness and maintaining answer stability, particularly in extractive question answering settings.

Chapter 6

Results

This chapter presents the experimental results of the proposed hallucination detection and mitigation framework across three benchmark datasets: TruthfulQA, FEVER, and Natural Questions (NQ). The results are organised to evaluate both detection performance and mitigation effectiveness under different question answering settings.

6.1 Overall Experimental Results

6.1.1 Overall Comparison Across Datasets

Table 15 summarises the overall performance across datasets, including hallucination rate, correctness, and improvement over the base model.

Table 15: Overall performance comparison across datasets.

Dataset	Base Halluc (%)	Final Halluc (%)	Δ Halluc	Final Correct (%)
TruthfulQA	88.3	89.0	+0.7	11.0
FEVER	59.5	29.5	-30.0	70.5
NQ	85.7	80.8	-4.8	19.2

As shown in Table 15, the proposed framework achieves varying performance across datasets. On FEVER, the hallucination rate decreases from 59.5% to 29.5%, corresponding to a reduction of 30.0%, while correctness reaches 70.5%. On Natural Questions, the hallucination rate decreases from 85.7% to 80.8%, corresponding to a reduction of 4.8%, with a final correctness of 19.2%. In contrast, on TruthfulQA, the hallucination rate increases slightly from 88.3% to 89.0%, while correctness remains at 11.0%.

Overall, the results show that the framework achieves substantial improvement on FEVER, moderate improvement on Natural Questions, and limited or negative improvement on TruthfulQA.

6.2 Hallucination Detection Results

6.2.1 Detection Results on TruthfulQA

Table 16: Hallucination detection performance on TruthfulQA.

Mode	Thr	F1 mean	F1 std	Q-std	A-std	ΔF1 vs Full
Full	0.70	0.9342	0.0006	0.0402	0.1106	0.0000
No Judge	0.73	0.9352	0.0005	0.0422	0.1254	+0.0010
No MNLI	0.59	0.9355	0.0005	0.0159	0.0511	+0.0014
No SelfCheck	0.69	0.9360	0.0002	0.0381	0.1070	+0.0019
Judge Only	0.60	0.9360	0.0002	0.0159	0.0543	+0.0019

The full model achieves a mean F1 score of 0.9342. Across ablation variants, performance differences are small, with several reduced configurations achieving slightly higher F1 scores (up to +0.0019). In terms of stability, the full model shows Q-std of 0.0402 and A-std of 0.1106, while some ablation variants exhibit lower variance.

6.2.2 Detection Results on FEVER

Table 17: Hallucination detection performance on FEVER.

Mode	Thr	F1 mean	F1 std	Q-std	A-std	ΔF1 vs Full
Full	0.20	0.9554	0.0027	0.0231	0.1697	0.0000
No Judge	0.31	0.7474	0.0224	0.0000	0.0372	-0.2080
No MNLI	0.37	0.9554	0.0027	0.0420	0.3086	0.0000
No SelfCheck	0.01	0.9554	0.0027	0.0289	0.2085	0.0000
Judge Only	0.01	0.9554	0.0027	0.0660	0.4766	0.0000

The full model achieves a mean F1 score of 0.9554. Removing the judge component leads to a decrease in F1 score to 0.7474. Other ablation variants maintain similar F1 scores but show increased variability. The full model exhibits Q-std of 0.0231 and A-std of 0.1697.

Table 18: Hallucination detection performance on Natural Questions.

Mode	Thr	F1 mean	F1 std	Q-std	A-std	ΔF1 vs Full
Full	0.77	0.9760	0.0060	0.0058	0.3226	0.0000
No Judge	0.76	0.9736	0.0073	0.0057	0.3317	-0.0024
No MNLI	0.69	0.9830	0.0070	0.0059	0.3013	+0.0070
No SelfCheck	0.81	0.9756	0.0027	0.0056	0.3190	-0.0004
Judge Only	0.79	0.9486	0.0046	0.0075	0.3367	-0.0274

6.2.3 Detection Results on Natural Questions

The full model achieves a mean F1 score of 0.9760. Performance across ablation settings remains relatively stable, with small variations in F1. The *no_mnli* variant achieves the highest F1 score of 0.9830. Stability metrics show low Q-std across all configurations and relatively higher A-std.

6.3 Mitigation Results

6.3.1 Mitigation Results on TruthfulQA

Table 19: Mitigation results on TruthfulQA (question-level)

Policy	Hallucination ↓	Accuracy ↑	Abstain	ΔScore
base	0.8827	0.1173	0.0020	-0.0000
rewrite_once	0.8860	0.1140	0.0020	0.0022
rewrite_K	0.9000	0.1000	0.0020	-0.0117
detector-cond (truthful)	0.8933	0.1067	0.0260	0.0033
detector-cond (utility)	0.8900	0.1100	0.0020	**0.0101**

The hallucination rate increases from 88.3% to 89.0%, while accuracy remains low at 11.0%. Different mitigation strategies show small variations in performance.

6.3.2 Mitigation Results on FEVER

Table 20: Mitigation results on FEVER (question-level)

Policy	Hallucination ↓	Accuracy ↑	Abstain	ΔScore
base	0.5950	0.4050	0.0000	0.0000
always_mild	0.5583	0.4417	0.0000	0.0169
always_strict_K	0.2967	0.7033	0.0000	0.1084
policy (ours)	**0.2950**	**0.7050**	0.0000	**0.1093**

The hallucination rate decreases from 59.5% to 29.5%, while accuracy increases to 70.5%. The proposed policy achieves the best overall performance among the evaluated strategies.

6.3.3 Mitigation Results on Natural Questions

Table 21: Mitigation results on Natural Questions (question-level)

Policy	Hallucination ↓	Accuracy ↑	EM	ΔScore
base	0.8567	0.1433	0.0983	0.0012
extract_once	0.8567	0.1433	0.0983	0.0012
extract_K	0.8567	0.1433	0.0983	0.0012
detector-conditioned (ours)	**0.8083**	**0.1917**	**0.1283**	**0.0741**

The hallucination rate decreases from 85.7% to 80.8%, while accuracy increases to 19.2%. The detector-conditioned strategy achieves the best performance across metrics.

6.4 Ablation Study

6.4.1 Stability Analysis

Tables 9–11 present stability results across datasets.

Across all datasets, the full model demonstrates low question-level variance and moderate answer-level variance. Ablation variants show increased variability in certain configurations.

6.5 Case Study Analysis

Representative case studies for each dataset are shown in Tables 12, 13, and 14. These examples illustrate improved, regressed, and error cases produced by the mitigation framework.

The results show that the framework can correct some incorrect answers, while also introducing regressions or failing to produce valid outputs in certain cases.

Chapter 7

Discussion

7.1 Cross-Dataset Discussion

7.1.1 Why FEVER Is Easier

FEVER represents a relatively structured fact verification setting, which makes hallucination detection and mitigation more effective.

Each claim is paired with explicit evidence sentences and a predefined label space (SUPPORTED, REFUTED, NOT ENOUGH INFO). This structured formulation reduces ambiguity and allows the model to directly ground its reasoning in the provided evidence.

In our framework, both detection and mitigation benefit significantly from this structure. The NLI-based factuality signal aligns well with the task formulation, as entailment and contradiction can be reliably computed between the evidence and the predicted label.

Furthermore, the mitigation process operates within a constrained space, as it enforces evidence-grounded rewriting and span extraction, ensuring that candidate answers are directly supported by the provided evidence.

As a result, FEVER exhibits the most consistent improvement across all mitigation strategies, with reduced hallucination rates and relatively low regression. This suggests that the proposed framework is particularly effective when the task provides strong grounding signals and limited output variability.

7.1.2 Why TruthfulQA Is More Challenging

In contrast, TruthfulQA presents a significantly more challenging setting for hallucination mitigation. Unlike FEVER, it is an open-ended question answering task designed to elicit common misconceptions and misleading responses from language models. Although reference answers are provided, they consist of multiple correct and incorrect responses rather than a single authoritative ground truth, making correctness inherently less well-defined.

This ambiguity reduces the effectiveness of both detection and mitigation components. The NLI-based factuality signal becomes less reliable, as there is no explicit evidence to serve as a grounding premise, and comparisons must instead be made against multiple reference answers with varying degrees of correctness. Consequently, the judge model relies more heavily on implicit world knowledge, which introduces additional uncertainty. Similarly, mitigation strategies based on rewriting and reranking may struggle to consistently identify truly correct answers among several plausible candidates.

Empirically, this leads to smaller improvements and higher variance compared to FEVER. The results suggest that hallucination mitigation in open-domain settings remains fundamentally challenging, particularly when correctness is underspecified and cannot be directly grounded in explicit evidence.

7.1.3 The Intermediate Behaviour of Natural Questions

Natural Questions (NQ) exhibits an intermediate behaviour between FEVER and TruthfulQA. Unlike FEVER, it does not provide explicit labels or structured evidence, but unlike TruthfulQA, it contains extractive answers grounded in real documents.

This enables mitigation strategies that leverage context-based span extraction to be effective in many cases, as the correct answer often appears explicitly within the retrieved context. However, the lack of structured supervision introduces ambiguity, particularly when multiple plausible spans exist or when the answer requires semantic abstraction beyond exact matching.

As observed in the case studies, the framework is able to correct factual errors when the correct answer is clearly present in the context, but may fail or regress when candidate spans are ambiguous or semantically similar. This reflects the limitations of extractive-guided reranking approaches, where the quality of candidate spans and scoring signals jointly determine the final outcome.

Overall, NQ demonstrates that hallucination mitigation benefits from grounding in real context, but still faces challenges due to ambiguity and variability in answer representation, placing it between the highly structured FEVER setting and the open-ended TruthfulQA scenario.

7.1.4 Detection-Mitigation Interaction

The interaction between detection and mitigation plays a critical role in the overall effectiveness of the framework. Rather than applying mitigation uniformly, the proposed approach uses the detector score to guide when and how to intervene, enabling a more adaptive and risk-aware correction strategy.

When the detector assigns a high confidence score to the base answer, the system preserves the original output, avoiding unnecessary modifications. For medium-confidence cases, controlled rewriting or reranking is applied, while low-confidence cases trigger more aggressive correction strategies, including multi-candidate generation and selection. In cases where no reliable candidate can be identified, the system may abstain, reflecting a conservative decision under high uncertainty.

This detector-conditioned design improves stability by reducing over-correction, which is a common issue in naive mitigation approaches. At the same time, it enables targeted intervention in cases where hallucination is more likely, while limiting unnecessary changes to already correct answers.

Experimental results show that this interaction leads to a more effective trade-off between correctness, abstention, and preservation, demonstrating the importance of integrating detection and mitigation within a unified framework.

7.1.5 Role of Abstention and Coverage Trade-offs

A key design consideration in hallucination mitigation is the trade-off between correctness and coverage. Allowing the system to abstain from answering can effectively reduce hallucination, but may also decrease the proportion of answered questions.

In the proposed framework, abstention is implemented as a safety mechanism when the highest-scoring candidate remains below a predefined confidence threshold. This is particularly beneficial in high-risk scenarios, where selecting any available answer would likely introduce incorrect information.

However, excessive abstention may reduce the practical usefulness of the system, especially in real-world applications where users expect a response. Therefore, the framework balances this trade-off by applying abstention selectively, using calibrated confidence thresholds derived from validation data.

The results indicate that a controlled use of abstention can significantly reduce hallucination without severely impacting coverage, highlighting its role as an important component in reliable question answering systems.

7.2 Discussion in Wider Context

As shown in Figure 5, the contribution of each detection component varies across datasets, indicating that the effectiveness of different signals is task-dependent rather than uniform. Overall, the multi-signal design provides a more robust detection framework than relying on any single component alone. For FEVER, removing the judge signal leads to the largest

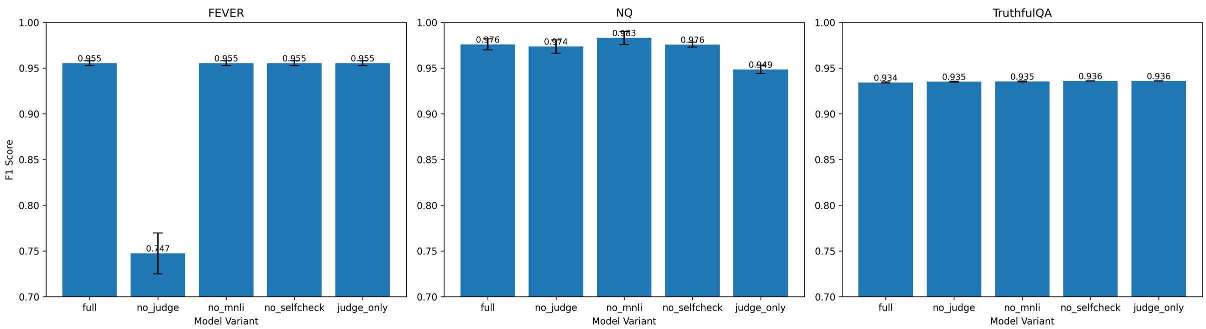

Figure 5: *Ablation study across datasets. Each subplot shows the impact of removing individual components on FEVER, NQ, and TruthfulQA respectively. Error bars indicate standard deviation across runs.*

performance drop, showing that semantic judgement plays a particularly important role in structured fact-verification settings. In contrast, the effects of removing individual components are smaller on Natural Questions and TruthfulQA, suggesting that the relative contribution of each signal becomes less distinct in more ambiguous or open-ended tasks.

Compared with SelfCheckGPT-style approaches, which rely solely on consistency across multiple generated answers, our method integrates multiple complementary signals, including LLM-as-a-judge, NLI-based factuality, and self-consistency. While SelfCheckGPT captures uncertainty through answer variability, it does not explicitly model factual correctness. In contrast, our approach combines semantic evaluation with contradiction detection, leading to more reliable and stable predictions.

Compared with single-model detection approaches, our framework demonstrates improved robustness by aggregating multiple perspectives. Single-model detectors are often sensitive to model bias and may fail in ambiguous cases. By combining judge-based reasoning, textual entailment, and consistency signals, the proposed method reduces reliance on any single source of information, resulting in more balanced and reliable hallucination estimation.

In terms of mitigation, naive prompting strategies such as direct rewriting or repeated generation often introduce additional noise, particularly in open-ended tasks. This is evident in the TruthfulQA results, where naive rewriting degrades performance. In contrast, the proposed detector-conditioned mitigation selectively applies correction based on estimated hallucination risk, avoiding unnecessary modifications and reducing regression errors.

Overall, these comparisons highlight that effective hallucination mitigation requires both accurate detection and controlled intervention. The results suggest that multi-signal approaches provide a more robust foundation for hallucination estimation, particularly when combined with adaptive, detector-guided mitigation strategies.

7.3 Summary of Findings

This chapter presents a comprehensive evaluation of the proposed hallucination detection and mitigation framework across three benchmark datasets: FEVER, TruthfulQA, and Natural Questions. The results reveal several key insights regarding the effectiveness of the proposed approach under different task settings.

First, the effectiveness of hallucination mitigation is strongly dependent on the availability of grounding information. The proposed framework achieves substantial improvements on FEVER, where explicit evidence and structured labels enable reliable correction. In contrast, performance gains are limited on Natural Questions and absent on TruthfulQA, highlighting the challenges of handling weakly grounded or open-ended tasks.

Second, the integration of detection and mitigation plays a critical role in improving system performance. The detector-conditioned strategy allows selective intervention based on confidence levels, reducing unnecessary modifications while targeting high-risk cases. This results in improved stability compared to naive, always-on mitigation approaches.

Third, the use of abstention introduces an important trade-off between correctness and coverage. By allowing the system to refrain from answering when confidence is low, hallucination can be reduced in high-risk scenarios. However, excessive abstention may limit the practical usability of the system, indicating the need for careful threshold calibration.

Despite these strengths, the proposed framework has limitations. In particular, its performance degrades in open-ended settings such as TruthfulQA, where the absence of reliable grounding signals makes both detection and mitigation more challenging. This suggests that future work should explore the integration of external knowledge sources or retrieval mechanisms to improve robustness in such scenarios.

Overall, the results demonstrate that effective hallucination mitigation requires both strong grounding signals and adaptive control mechanisms. The proposed multi-signal framework provides a robust foundation for hallucination estimation, particularly in structured and semi-structured question answering tasks.

Chapter 8

Project Management

8.1 Project Planning and Timeline

The project was planned and managed using a structured timeline, as illustrated in Figure 6. The overall workflow was divided into several phases, including initial research, literature review, data preparation, development of detection mechanism, development of mitigation strategies, evaluation, dissertation writing, and Q&A session. This phased design ensured a logical progression from foundational understanding to system development and final evaluation.

In the early stages (September to December), the focus was on foundational tasks, including project proposal development, literature review, and dataset selection. This phase established the theoretical background and ensured that appropriate datasets and models were selected for subsequent experiments. In particular, key concepts related to hallucination detection and mitigation were reviewed, forming the basis for the design of the proposed framework.

From December to January, the project progressed to data preparation and baseline implementation, followed by the development of the hallucination detection mechanism. This stage included preprocessing datasets, generating baseline model outputs, and implementing core detection components such as LLM-as-a-judge evaluation, NLI-based factual verification, and self-consistency analysis.

In the later stages (January to March), the project focused on integrating detection and mitigation into a closed-loop framework, as well as developing prompt-based mitigation strategies. This included candidate generation, filtering, and reranking mechanisms, forming the core contribution of the project.

Figure 6: *Project timeline and task scheduling (Gantt chart).*

Subsequently, evaluation and benchmarking were conducted from March to April, including performance evaluation of both detection and mitigation components, ablation studies to assess the contribution of individual signals, and cross-dataset analysis across TruthfulQA, FEVER, and Natural Questions. These experiments ensured a comprehensive assessment of the proposed framework in different task settings.

Finally, the last phase (April to May) was dedicated to dissertation writing, system demonstration, and preparation for the final presentation and Q&A session. This stage involved consolidating experimental results, refining the written report, and preparing materials to clearly communicate the project contributions.

Overall, the project timeline reflects a structured and iterative development process, where system design, implementation, and evaluation were closely aligned to ensure both technical completeness and experimental rigor.

8.2 Task Allocation and Milestones

To ensure that the project progressed in a structured and manageable way, the overall work was divided into several task categories, each corresponding to a major component of the proposed framework. In addition, a set of milestones was defined to monitor progress and ensure that the project remained aligned with its objectives.

8.2.1 Task Allocation

The project tasks were organised into six main categories.

First, an initial research and literature review stage was carried out to establish the theoretical background of hallucination in large language models. This task included reviewing prior work on hallucination detection, estimation, and mitigation, as well as identifying the research gap addressed in this project.

Second, a methodology and system design task was defined, focusing on the formulation of the multi-signal framework and the overall closed-loop pipeline. This stage translated the insights from the literature review into a concrete technical design.

Third, a data preparation task was carried out to process the selected benchmark datasets, including TruthfulQA, FEVER, and Natural Questions. This included subset construction, preprocessing, and reference extraction where necessary.

Fourth, the system implementation stage was divided into two main technical tasks: hallucination detection and mitigation. The detection task focused on implementing the multi-signal detector, including LLM-as-a-judge, NLI-based factuality, and self-consistency components. The mitigation task focused on implementing the detector-conditioned correction framework, including candidate generation, filtering, reranking, and dataset-specific adaptations.

Fifth, an evaluation task was defined to assess both detection and mitigation performance. This included threshold tuning, Monte Carlo experiments, ablation studies, and cross-dataset comparisons.

Finally, a dissertation writing and presentation preparation task was carried out in parallel with the later technical stages. This ensured that implementation progress, experimental results, and analysis could be continuously documented and refined throughout the project.

8.2.2 Key Milestones

Several key milestones were defined to monitor project progress.

The first milestone was the completion of the literature review and problem definition. This marked the point at which the project scope, research gap, and objectives were clearly established.

The second milestone was the completion of dataset preparation and the initial detection prototype. At this stage, the project had moved from conceptual planning to practical implementation.

The third milestone was the completion of the closed-loop integration between hallucination detection and mitigation. This represented the implementation of the core system contribution.

The fourth milestone was the completion of the mitigation strategy and dataset-specific adaptations. This ensured that the framework could be evaluated across multiple task settings.

The fifth milestone was the completion of evaluation experiments, including ablation studies and cross-dataset analysis. This milestone provided the evidence required to assess the effectiveness and limitations of the proposed framework.

The final milestone was the completion of the dissertation, demonstration materials, and Q&A preparation. This stage marked the transition from technical development to final reporting and presentation.

Overall, the task allocation and milestone planning helped ensure that the project remained organised and goal-oriented. By breaking the work into clearly defined components and monitoring progress through milestone completion, the project was able to maintain a logical development process from initial research to final evaluation and submission.

8.3 Challenges and Risk Management

Throughout the project, several practical and technical challenges were encountered. These challenges affected not only the implementation of the proposed framework, but also the planning and management of the overall project. To reduce their impact, a number of risk management strategies were adopted.

8.3.1 Implementation and Experimental Challenges

One major challenge was the computational cost of the proposed framework. The system integrates multiple components, including answer generation, LLM-as-a-judge evaluation, NLI-based factual verification, and self-consistency analysis. Running these components across three datasets and multiple experimental settings significantly increased runtime and computational overhead.

A second challenge was the stochastic nature of large language model outputs. Even under controlled decoding settings, generated answers could vary across runs, which affected the reproducibility of both detection and mitigation results. This made it difficult to rely on single-run performance as a stable indicator.

Another challenge arose from the heterogeneity of the selected datasets. TruthfulQA, FEVER, and Natural Questions differ substantially in task formulation, grounding information, and answer structure. As a result, a single mitigation strategy could not be applied uniformly without adjustment.

In addition, the mitigation performance was not equally effective in all datasets. In particular, open-ended tasks such as TruthfulQA proved much more difficult, as the absence of explicit grounding made correction less reliable. This introduced a project risk in terms of achieving consistent performance improvements.

Finally, time management was itself a practical challenge. The project required balancing system development, experimentation, result analysis, and dissertation writing within a limited timeframe.

8.3.2 Risk Management Strategies

To manage computational cost, answer caching was introduced so that previously generated outputs could be reused across experiments. This reduced redundant model calls and improved efficiency during repeated evaluations.

To address reproducibility issues, a Monte Carlo evaluation protocol was adopted. Instead of relying on a single run, multiple runs were conducted and results were reported using mean and standard deviation. This provided a more reliable estimate of performance and reduced the impact of output variability.

To manage dataset heterogeneity, the project adopted a unified high-level framework with dataset-specific adaptations. The overall detection–mitigation pipeline remained consistent, while task-specific adjustments were introduced for FEVER, TruthfulQA, and Natural Questions to ensure compatibility with different task structures.

To reduce the risk associated with limited mitigation performance in open-ended tasks, the project scope was managed carefully. Rather than assuming uniform success across all datasets, the evaluation was designed to compare performance under different levels of grounding. This allowed the project to generate meaningful findings even when mitigation gains were limited in certain settings.

From a project planning perspective, milestone-based management was used to keep the work on track. Early stages focused on literature review and framework design, while later stages were reserved for implementation, evaluation, and writing. Dissertation writing was also started before the completion of all experiments, reducing the risk of excessive workload near the deadline.

Overall, these risk management strategies improved the efficiency, robustness, and feasibility of the project. They also ensured that the project could continue progressing despite technical uncertainty and experimental variability.

8.4 Project Adjustments

Although the project was initially planned using a structured timeline, several adjustments were made during the development process in response to practical challenges and experimental findings. These adjustments were necessary to ensure that the project remained feasible, robust, and aligned with its research objectives.

One major adjustment concerned the design of the mitigation strategy. The original plan aimed to develop a unified mitigation approach that could be applied consistently across all datasets. However, during experimentation, it became clear that the three datasets (TruthfulQA, FEVER, and Natural Questions) differ significantly in task formulation, grounding conditions, and answer structure. As a result, a single mitigation strategy could not be effectively applied without modification. To address this, the framework was adjusted to adopt a dataset-specific mitigation design, while maintaining a unified overall architecture. This allowed the system to better adapt to different task requirements and improved overall effectiveness.

A second important adjustment was related to the evaluation methodology. Initially, the project considered evaluating model performance based on single-run experiments. However, due to the stochastic nature of large language models, the results were found to vary across runs, leading to instability in performance measurements. To mitigate this issue, a Monte Carlo evaluation protocol was introduced, where multiple runs were conducted and results were reported using mean and standard deviation. This adjustment significantly improved the reliability and robustness of the evaluation.

Another adjustment involved refining the project objectives for mitigation. The initial goal was to achieve consistent hallucination reduction across all datasets. However, experimental results showed that mitigation performance varied depending on the level of grounding available in each dataset. In particular, open-ended tasks such as TruthfulQA were more difficult to improve. As a result, the focus of the project was adjusted from achieving uniform improvement to analysing how mitigation effectiveness differs across datasets with varying levels of grounding. This allowed the project to produce more meaningful and realistic insights.

In addition to technical adjustments, minor changes were also made to project scheduling. While the original plan allocated most of the dissertation writing to the final stage, it became necessary to start writing earlier in parallel with experimentation. This helped reduce time pressure towards the end of the project and ensured that results and analysis could be documented progressively.

Overall, these adjustments reflect an adaptive development process, where both technical design and project planning were refined based on empirical observations. This flexibility was essential in ensuring that the project remained both achievable and academically rigorous.

8.5 Reflection

This project provided valuable insights into the importance of effective project management in research-oriented system development. Beyond the technical implementation, the project required careful planning, continuous adjustment, and the ability to manage uncertainty arising from both experimental and methodological factors.

One key lesson learned is the importance of structuring the project into well-defined phases with clear milestones. The use of a phased timeline helped ensure that foundational tasks, such as literature review and system design, were completed before moving on to implementation and evaluation. This contributed to a more organised and manageable workflow throughout the project.

Another important takeaway is the need to design systems in a flexible and adaptive manner. Initially, the project aimed to develop a unified mitigation strategy applicable across all datasets. However, it became evident that differences in task formulation and grounding conditions required dataset-specific adaptations. This highlighted the importance of iterative refinement and responsiveness to empirical findings, rather than strictly adhering to the original plan.

The project also emphasised the challenges of working with stochastic models such as large language models. The variability of model outputs made it necessary to adopt robust evaluation strategies, such as Monte Carlo experiments, to ensure reliable and reproducible results. This experience reinforced the importance of incorporating stability considerations into both experimental design and project planning.

In terms of limitations, some aspects of project planning could have been improved. In particular, the complexity of mitigation in open-ended tasks such as TruthfulQA was underestimated in the early stages. As a result, more time than expected was required to analyse and interpret experimental results. Additionally, balancing system development and dissertation writing proved challenging, especially in the later stages of the project.

If the project were to be conducted again, several improvements could be made. First, more extensive pilot experiments could be conducted at an earlier stage to better understand dataset-specific challenges and guide design decisions. Second, a more modular system design could be adopted from the outset to facilitate easier adaptation across different tasks. Finally, initiating evaluation and writing activities earlier in parallel with development would help reduce time pressure and improve overall efficiency.

Overall, this project demonstrates that successful system development requires not only technical expertise but also effective project management. The ability to plan, adapt, and critically reflect on both successes and limitations is essential for conducting rigorous and impactful research.

Chapter 9

Reflection and Future Work

9.1 Summary of Contributions

This project proposes a multi-signal framework for hallucination detection, estimation, and mitigation in large language models. The framework integrates multiple complementary signals, including LLM-as-a-judge, NLI-based factual consistency, self-consistency, and auxiliary signals, to provide a more robust estimation of hallucination risk compared to single-signal approaches.

A key contribution of this work is the design of a closed-loop pipeline, where hallucination detection is directly integrated with mitigation. The estimated reliability score is used to guide detector-conditioned mitigation strategies, enabling adaptive intervention based on confidence levels. This allows the system to preserve high-confidence answers while selectively correcting low-confidence ones, thereby reducing unnecessary modifications while focusing on high-risk cases.

Furthermore, the proposed framework introduces a unified scoring mechanism that aggregates heterogeneous signals into a continuous reliability score. This score supports both binary hallucination classification and fine-grained risk estimation, providing a principled basis for threshold-based decision-making and mitigation control.

The framework is evaluated across three benchmark datasets, namely TruthfulQA, FEVER, and Natural Questions, covering diverse settings including open-ended generation, evidence-based verification, and extractive question answering. Experimental results demonstrate that the proposed approach achieves strong performance in structured, evidence-grounded scenarios and improves robustness through multi-signal integration, while highlighting the challenges of hallucination mitigation in open-domain settings.

Overall, this project contributes a unified, prompt-engineering-based and closed-loop solution for hallucination detection and mitigation, emphasising the importance of combining complementary signals and adaptive control strategies to improve the reliability and stability of large language models across diverse tasks.

9.2 Critical Reflection

While the proposed framework demonstrates promising results, several important observations can be made regarding its effectiveness and design choices.

First, the framework performs significantly better on structured datasets such as FEVER, where explicit evidence provides strong grounding for both detection and mitigation. This confirms that NLI-based reasoning and constraint-based filtering are highly effective when reliable external information is available. In particular, the use of entailment–contradiction scoring and contradiction-aware filtering in the NLI module plays a critical role in eliminating unsupported or conflicting candidate answers.

However, performance on open-ended datasets such as TruthfulQA remains relatively limited. In these settings, the absence of explicit grounding introduces substantial ambiguity, making it difficult for both the detector and the mitigation strategy to reliably distinguish correct answers from plausible but incorrect alternatives. Since the framework relies primarily on semantic judgment (LLM-as-a-judge) and self-consistency signals, the detector may assign high confidence to internally consistent but factually incorrect answers. This reflects a key limitation that consistency does not necessarily imply factual correctness. This highlights a fundamental limitation of post-hoc mitigation approaches when operating without external knowledge.

Another key observation is the strong dependency on candidate quality. In many failure cases, the mitigation strategy does not fail due to incorrect scoring, but rather because the correct answer is not present in the candidate pool. This is especially evident in the candidate generation stage, where prompt-based rewriting or extractive heuristics may fail to produce factually correct alternatives, limiting the effectiveness of downstream filtering and reranking. This indicates that the overall system performance is bounded not only by the detection model, but also by the quality and diversity of the generated candidates.

Furthermore, the effectiveness of the detector-conditioned policy depends on the reliability of the aggregated score. In cases where different signals (e.g., judge, NLI, and self-consistency) produce conflicting indications, the weighted aggregation may lead to unstable or suboptimal decisions. This suggests that more advanced aggregation or calibration strategies may be required to better balance competing signals.

From a system design perspective, the integration of multiple signals improves robustness and stability, as different components capture complementary aspects of hallucination. However, this also increases system complexity and computational cost, requiring multiple model calls and additional processing steps, including repeated LLM inference, NLI evaluation, and SelfCheck-based consistency estimation across multiple sampled answers. This trade-off between performance and efficiency represents an important consideration for practical deployment.

9.3 Limitations

Despite the promising results, several limitations of this work should be acknowledged.

First, the framework relies on pre-generated candidate answers, which limits its ability to correct errors when no correct candidate is available. As a result, the mitigation process is inherently constrained by the quality of the initial generation step. In particular, candidate answers are produced using prompt-based rewriting or extractive heuristics, which do not guarantee coverage of the correct answer space. When the correct answer is absent from the candidate pool, the subsequent filtering and reranking stages cannot recover it, regardless of the effectiveness of the detection model.

Second, the effectiveness of the detection model is highly task-dependent. While strong performance is achieved on structured datasets such as FEVER, improvements are limited on open-ended tasks such as TruthfulQA, where ground truth is less well-defined and explicit evidence is unavailable.

In particular, the framework exhibits reduced effectiveness in open-domain question answering settings. In such scenarios, the absence of explicit grounding information makes it inherently difficult to reliably distinguish between correct answers and plausible but incorrect responses. Unlike structured datasets such as FEVER, where external evidence provides a reliable reference for verification, open-domain tasks rely heavily on implicit world knowledge and model-generated candidates.

As a result, the detector may assign high confidence scores to semantically fluent but factually incorrect answers, especially when LLM-as-a-judge and self-consistency signals agree. This leads to error propagation into the mitigation stage, where incorrect candidates may be preserved or even preferred during reranking.

Third, the use of LLM-based evaluation introduces inherent uncertainty. The judge model itself may exhibit bias, inconsistency, or overconfidence, which can propagate errors into both detection and mitigation stages. In this project, the judge component relies on an instruction-tuned LLM, which is sensitive to prompt design and may produce unstable judgments across different inputs or sampling conditions.

Furthermore, the aggregation of multiple signals introduces additional complexity. Since different signals (e.g., LLM-as-a-judge, NLI-based factuality, and self-consistency) may produce conflicting indications, the weighted combination used in the scoring function may not always reflect the true factual correctness of the answer. This limitation highlights the challenge of designing reliable multi-signal aggregation mechanisms.

Finally, the computational cost of the proposed framework is relatively high. Multi-signal aggregation, multi-sample generation, and repeated model evaluation significantly increase runtime, which may limit scalability and real-world applicability. In particular, the use of multiple sampled answers for self-consistency estimation, repeated NLI inference for entailment and contradiction scoring, and iterative candidate generation in the mitigation stage all contribute to increased computational overhead.

Overall, these limitations highlight that hallucination mitigation remains a fundamentally challenging problem, particularly in open-domain settings where reliable grounding signals are unavailable.

9.4 Future Work

Several directions can be explored to further improve the proposed framework.

First, future work could integrate retrieval-augmented generation (RAG) to provide external knowledge grounding, particularly for open-domain tasks such as TruthfulQA. Incorporating reliable evidence would address the current limitation where the framework relies primarily on internal signals, and is therefore prone to selecting semantically plausible but factually incorrect answers. By combining retrieval with the existing multi-signal detector, it would be possible to further improve both detection accuracy and mitigation effectiveness.

Second, improving the quality and diversity of candidate answers represents an important direction. Since the current mitigation framework is constrained by the candidate pool, future work could explore more advanced generation strategies, such as guided decoding, iterative refinement, or diversity-aware sampling techniques. These approaches could increase the likelihood of including correct answers in the candidate set, thereby enhancing the effectiveness of downstream filtering and reranking.

Third, future research could investigate more advanced multi-signal aggregation and calibration methods. In the current framework, signals are combined using manually designed weighting schemes, which may not fully capture complex interactions between signals. Learning-based aggregation methods, such as trainable scoring models or uncertainty-aware calibration techniques, could provide more reliable and adaptive estimation of hallucination risk.

Fourth, improving the efficiency of the framework is an important practical direction. The current system requires multiple model calls, including repeated LLM inference, NLI evaluation, and self-consistency estimation, which introduces significant computational overhead. Future work could explore lightweight approximations, signal pruning strategies, or parallelised implementations to reduce runtime while maintaining performance.

Finally, the framework could be extended to real-world applications, such as conversational agents or decision-support systems. In such settings, balancing reliability, efficiency, and user experience will be critical. In addition, incorporating user feedback or interactive correction mechanisms could enable the system to continuously improve its performance and better adapt to real-world usage scenarios.

Chapter 10

Conclusion

This project investigated the problem of hallucination detection and mitigation in large language models and proposed a unified multi-signal and closed-loop framework. The approach integrates complementary signals, including LLM-as-a-judge, NLI-based factual consistency, self-consistency, and auxiliary signals, to estimate the reliability of generated answers and guide adaptive mitigation strategies.

A key contribution of this work is the design of a detector-conditioned pipeline, where hallucination detection is not treated as a standalone task, but is directly integrated with answer correction. By using a unified reliability score to control candidate filtering, reranking, and intervention strength, the framework enables more effective and interpretable mitigation compared to conventional post-hoc approaches.

Experimental evaluation across three benchmark datasets (TruthfulQA, FEVER, and Natural Questions) demonstrates that the proposed framework achieves strong performance in structured, evidence-grounded settings, particularly in FEVER, where explicit evidence enables effective NLI-based verification and filtering. In contrast, improvements in open-ended settings such as TruthfulQA remain limited, highlighting the challenges of hallucination mitigation in the absence of reliable grounding information. These results indicate that the effectiveness of hallucination mitigation is highly dependent on the availability of external evidence and the quality of generated candidates.

Overall, this project shows that combining multiple complementary signals within a closed-loop framework is a promising direction for improving the reliability of large language models. At the same time, it highlights that hallucination remains a fundamentally challenging problem, particularly in open-domain scenarios, and further research is required to develop more robust, efficient, and generalisable solutions.

References

[1] O. N. Oyelade, H. Wang, and K. Rafferty, "A survey of adaptation of large language models to idea and hypothesis generation: Downstream task adaptation, knowledge distillation approaches and challenges," *ACM Comput. Surv.*, vol. 58, no. 6, Dec. 2025, ISSN: 0360-0300. DOI: 10.1145/3774628. [Online]. Available: https://doi.org/10.1145/3774628.

[2] X. Chen, J. Xiang, S. Lu, Y. Liu, M. He, and D. Shi, *Evaluating large language models in medical applications: A survey*, 2024. arXiv: 2405.07468 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2405.07468.

[3] A. Alansari and H. Luqman, *Large language models hallucination: A comprehensive survey*, 2026. arXiv: 2510.06265 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2510.06265.

[4] Z. Ji, N. Lee, R. Frieske, *et al.*, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, Mar. 2023, ISSN: 1557-7341. DOI: 10.1145/3571730. [Online]. Available: http://dx.doi.org/10.1145/3571730.

[5] X. Lin, Y. Ning, J. Zhang, *et al.*, *Llm-based agents suffer from hallucinations: A survey of taxonomy, methods, and directions*, 2025. arXiv: 2509.18970 [cs.AI]. [Online]. Available: https://arxiv.org/abs/2509.18970.

[6] C. Foo, P. S. Foong, C. Nadal, N. Ureyang, T. Naylin, and G. C. H. Koh, "The benefits and risks of llms for facilitating medical decision-making among laypersons," in *Proceedings of the 2025 ACM Designing Interactive Systems Conference*, ser. DIS '25, Association for Computing Machinery, 2025, pp. 3173–3191, ISBN: 9798400714856. DOI: 10.1145/3715336.3735779. [Online]. Available: https://doi.org/10.1145/3715336.3735779.

[7] N. Dziri, S. Milton, M. Yu, O. Zaiane, and S. Reddy, "On the origin of hallucinations in conversational models: Is it the datasets or the models?" In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, M. Carpuat, M.-C. de Marneffe, and I. V. Meza Ruiz, Eds., Seattle, United States: Association for Computational Linguistics, Jul. 2022, pp. 5271–5285. DOI: 10.18653/v1/2022.naacl-main.387. [Online]. Available: https://aclanthology.org/2022.naacl-main.387/.

[8] S. Kadavath, T. Conerly, A. Askell, *et al.*, *Language models (mostly) know what they know*, 2022. arXiv: 2207.05221 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2207.05221.

[9] P. Lewis, E. Perez, A. Piktus, *et al.*, *Retrieval-augmented generation for knowledge-intensive nlp tasks*, 2021. arXiv: 2005.11401 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2005.11401.

[10] P. Manakul, A. Liusie, and M. J. F. Gales, *Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models*, 2023. arXiv: 2303.08896 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2303.08896.

[11] L. Huang, W. Yu, W. Ma, *et al.*, "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," *ACM Transactions on Information Systems*, vol. 43, no. 2, pp. 1–55, Jan. 2025, ISSN: 1558-2868. DOI: 10.1145/3703155. [Online]. Available: http://dx.doi.org/10.1145/3703155.

[12] Y. Bang, Z. Ji, A. Schelten, *et al.*, *Hallulens: Llm hallucination benchmark*, 2025. arXiv: 2504.17550 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2504.17550.

[13] K. Shuster, S. Poff, M. Chen, D. Kiela, and J. Weston, *Retrieval augmentation reduces hallucination in conversation*, 2021. arXiv: 2104.07567 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2104.07567.

[14] P. Chhikara, *Mind the confidence gap: Overconfidence, calibration, and distractor effects in large language models*, 2025. arXiv: 2502.11028 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2502.11028.

[15] M. Cossio, *A comprehensive taxonomy of hallucinations in large language models*, 2025. arXiv: 2508.01781 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2508.01781.

[16] K. Guu, K. Lee, Z. Tung, P. Pasupat, and M.-W. Chang, *Realm: Retrieval-augmented language model pre-training*, 2020. arXiv: 2002.08909 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2002.08909.

[17] L. Tu, R. Meng, S. Joty, Y. Zhou, and S. Yavuz, *Investigating factuality in long-form text generation: The roles of self-known and self-unknown*, 2025. arXiv: 2411.15993 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2411.15993.

[18] S. Kang, Y. F. Bakman, D. N. Yaldiz, B. Buyukates, and S. Avestimehr, *Uncertainty quantification for hallucination detection in large language models: Foundations, methodology, and future directions*, 2025. arXiv: 2510.12040 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2510.12040.

[19] A. T. Kalai, O. Nachum, S. S. Vempala, and E. Zhang, *Why language models hallucinate*, 2025. arXiv: 2509.04664 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2509.04664.

[20] X. Wang, J. Wei, D. Schuurmans, *et al.*, *Self-consistency improves chain of thought reasoning in language models*, 2023. arXiv: 2203.11171 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2203.11171.

[21] S. Lin, J. Hilton, and O. Evans, *Truthfulqa: Measuring how models mimic human falsehoods*, 2022. arXiv: 2109.07958 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2109.07958.

[22] S. Longpre, K. Perisetla, A. Chen, N. Ramesh, C. DuBois, and S. Singh, *Entity-based knowledge conflicts in question answering*, 2022. arXiv: 2109.05052 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2109.05052.

[23] O. Honovich, R. Aharoni, J. Herzig, *et al.*, *True: Re-evaluating factual consistency evaluation*, 2022. arXiv: 2204.04991 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2204.04991.

[24] A. Williams, N. Nangia, and S. R. Bowman, *A broad-coverage challenge corpus for sentence understanding through inference*, 2018. arXiv: 1704.05426 [cs.CL]. [Online]. Available: https://arxiv.org/abs/1704.05426.

[25] P. Roit, J. Ferret, L. Shani, *et al.*, "Factually consistent summarization via reinforcement learning with textual entailment feedback," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds., Toronto, Canada: Association for Computational Linguistics, Jul. 2023, pp. 6252–6272. DOI: 10.18653/v1/2023.acl-long.344. [Online]. Available: https://aclanthology.org/2023.acl-long.344/.

[26] P. Laban, T. Schnabel, P. N. Bennett, and M. A. Hearst, *Summac: Re-visiting nli-based models for inconsistency detection in summarization*, 2021. arXiv: 2111.09525 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2111.09525.

[27] G. Izacard and E. Grave, *Leveraging passage retrieval with generative models for open domain question answering*, 2021. arXiv: 2007.01282 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2007.01282.

[28] L. Zheng, W.-L. Chiang, Y. Sheng, *et al.*, *Judging llm-as-a-judge with mt-bench and chatbot arena*, 2023. arXiv: 2306.05685 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2306.05685.

[29] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu, *G-eval: Nlg evaluation using gpt-4 with better human alignment*, 2023. arXiv: 2303.16634 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2303.16634.

[30] P. Wang, L. Li, L. Chen, *et al.*, *Large language models are not fair evaluators*, 2023. arXiv: 2305.17926 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2305.17926.

[31] J. Wang, X. Hu, W. Hou, *et al.*, *On the robustness of chatgpt: An adversarial and out-of-distribution perspective*, 2023. arXiv: 2302.12095 [cs.AI]. [Online]. Available: https://arxiv.org/abs/2302.12095.

[32] L. Ouyang, J. Wu, X. Jiang, *et al.*, *Training language models to follow instructions with human feedback*, 2022. arXiv: 2203.02155 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2203.02155.

[33] J. Wei, X. Wang, D. Schuurmans, *et al.*, *Chain-of-thought prompting elicits reasoning in large language models*, 2023. arXiv: 2201.11903 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2201.11903.

[34] V. Karpukhin, B. Oğuz, S. Min, *et al.*, *Dense passage retrieval for open-domain question answering*, 2020. arXiv: 2004.04906 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2004.04906.

[35] A. Madaan, N. Tandon, P. Gupta, *et al.*, *Self-refine: Iterative refinement with self-feedback*, 2023. arXiv: 2303.17651 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2303.17651.

[36] N. Shinn, F. Cassano, E. Berman, A. Gopinath, K. Narasimhan, and S. Yao, *Reflexion: Language agents with verbal reinforcement learning*, 2023. arXiv: 2303.11366 [cs.AI]. [Online]. Available: https://arxiv.org/abs/2303.11366.

[37] Y. Bai, S. Kadavath, S. Kundu, *et al.*, *Constitutional ai: Harmlessness from ai feedback*, 2022. arXiv: 2212.08073 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2212.08073.

Appendix A

Prompt Templates

This section presents the prompt templates used in the proposed framework for answer generation, hallucination detection, and mitigation. These prompts are designed to enforce structured outputs and guide the behaviour of the language model without requiring fine-tuning.

1.1 Base Generation Prompt

A general prompt template is used across all datasets to ensure structured output:

> **General Base Prompt**
>
> You are a knowledgeable assistant. Answer the following question concisely and accurately.
> Question: {question}
> Provide your final answer in the format: FINAL_ANSWER: your answer

However, dataset-specific constraints are applied to align with task requirements:

1.1.1 TruthfulQA (Generation Setting)

> **TruthfulQA Generation Prompt**
>
> Question: {question}
> Answer briefly and directly.
> Output exactly: FINAL_ANSWER: answer

In addition to base generation, multiple prompt templates are used for answer refinement during mitigation. These include a mild editing prompt that preserves original wording while correcting errors, and a strict correction prompt that removes misconceptions and allows abstention ("I do not know") when uncertainty is high.

1.1.2 FEVER (Verification Setting)

FEVER Prompt

Determine whether the claim is SUPPORTED, REFUTED, or NOT ENOUGH INFO based on the given evidence.

Claim: claim

Evidence: evidence

You MUST output in this format:

- LABEL: SUPPORTED / REFUTED / NOT ENOUGH INFO
- REASON: one short sentence based ONLY on the evidence

If the evidence is insufficient, choose NOT ENOUGH INFO.

1.1.3 Natural Questions (Extractive Setting)

NQ Prompt

You are a short-answer QA system.

Question: {question}

Context: {context}

Rules:

- Answer using only information from the Context
- Return the SHORTEST correct short answer
- Prefer 1-3 words
- Do NOT return a sentence
- Do NOT explain
- Do NOT include extra context

Output exactly one line:

FINAL_ANSWER:

Although the prompt encourages short, context-grounded answers, it does not explicitly require the output to be an exact contiguous span from the context. In the implementation, extractive behaviour is further enforced during post-processing through span-based candidate generation, answer compression, context alignment, candidate filtering, and detector-based reranking. Specifically, candidate spans are generated from the context using token-overlap heuristics, validated against the context, and combined with a small number of short rewritten answers under relaxed constraints. Therefore, the Natural Questions setting is implemented as an extractive-dominant hybrid pipeline rather than a strictly prompt-constrained extractive QA system.

1.2 LLM-as-a-Judge Prompt

To provide a semantic-level evaluation signal, we employ a Large Language Model (LLM) as an external factuality judge. This component evaluates whether a generated answer is correct with respect to task-specific references, capturing semantic equivalence beyond surface-level matching.

Judge Prompt

You are a strict factuality evaluator.

- Minor wording differences are acceptable

- Partial match is acceptable if core meaning matches

Mark True if:

- The model answer is semantically correct with respect to the reference,

- AND it contains no false or contradictory statements.

Extra correct information is allowed and should NOT cause False.

Mark False if:

- the answer fails to capture the reference meaning,

- OR contains any factual error or contradiction.

Input:

Question / Claim: {question}

Reference: {reference}

Model Answer: {answer}

Respond with exactly one token: True or False.

The judge is implemented using a separate instruction-tuned LLM (`qwen2.5:7b-instruct`) to avoid interference with the generation model. The output is parsed as a binary signal, where `True` indicates that the model output is factually correct with respect to the reference under the task-specific evaluation criteria, and `False` indicates a hallucinated or incorrect output.

In the implementation, the judge output is converted into a scalar score:

$$\text{Score}_{\text{judge}} = \begin{cases} w_{\text{judge}} & \text{if judge=True} \\ 0.0 & \text{if judge=False} \end{cases}$$

where w_{judge} is a task-dependent weight (e.g., 0.8 or 1.0) that reflects the relative importance of the judge signal in the overall scoring function.

This scoring scheme directly corresponds to the implementation, where a positive judge decision contributes a fixed value to the overall hallucination score, while negative decisions contribute zero.

Dataset-specific Adaptation

Although a unified judge framework is used, the evaluation logic is adapted to match each dataset's task formulation.

TruthfulQA

For open-ended truthfulness evaluation, answers are assessed using a multi-signal framework that integrates LLM-based judgment, natural language inference (NLI), and lexical matching. The judge evaluates whether the response avoids common misconceptions and follows truthful reasoning patterns.

Crucially, NLI is applied to ensure the generated answer is supported by correct references while not aligning with known incorrect answers, thereby explicitly penalizing misleading or false content. Unlike NQ, semantic similarity alone is insufficient; factual correctness is strictly enforced with both positive and negative evidence constraints.

FEVER

For fact verification, correctness is determined by both label matching and evidence consistency. Specifically, the predicted label must match the ground-truth label, and the answer must not contradict the provided evidence.

This corresponds to the implementation:

```
fever_gold_ok(answer, gold_label, evidence)
```

where label correctness is enforced by exact matching, and evidence consistency is verified using a separate NLI model to detect contradictions.
Additionally, an LLM-based judge:

```
judge_truth(question, answer, gold_label, evidence)
```

is used as an auxiliary signal in the overall hallucination scoring, rather than as the sole correctness criterion.

Natural Questions (NQ)
For extractive question answering, the judge evaluates whether the generated answer preserves the semantic meaning of the reference short answer while ensuring that no false or contradictory information is introduced. This corresponds to the implementation:

```
judge_truth(question, answer, reference)
```

where correctness is determined by both semantic equivalence and the absence of factual errors, rather than exact string matching.

Across all datasets, the judge operates as a task-agnostic semantic evaluator, while task-specific constraints are injected through prompt conditioning and input structure. This design ensures consistency of the evaluation signal while maintaining compatibility with heterogeneous benchmarks.

The use of LLM-as-a-Judge allows the system to capture semantic correctness that cannot be reliably measured using exact match or token-level overlap alone, especially for paraphrased or partially matching answers.

1.3 Mitigation Prompts

To reduce hallucinations in generated answers, a prompt-engineering-based mitigation strategy is applied in a closed-loop framework. Unlike simple one-shot prompting, the system dynamically selects different rewriting prompts based on the estimated hallucination risk, forming a detector-conditioned correction pipeline.

1.3.1 Overall Strategy

Given an initial answer, the system first computes a calibrated correctness score using the hallucination detector. Based on this score, three levels of mitigation are applied:

1. low-risk answers are kept unchanged
2. medium-risk answers undergo mild rewriting
3. high-risk answers are corrected using stricter prompts with multiple candidate generations and reranking

This design ensures that corrections are only applied when necessary, reducing unnecessary modifications to already correct answers.

1.3.2 Mild Correction Prompt

For moderately risky answers, a conservative editing prompt is used:

- preserve as much of the original wording as possible,
- modify only incorrect or misleading parts,
- avoid introducing new information.

This corresponds to a minimal intervention strategy, aiming to fix local factual errors without altering the overall structure of the answer.

1.3.3 Strict Correction Prompt

For high-risk answers, a stronger prompt is applied to enforce factual correctness:

- remove false or speculative content,
- explicitly correct misconceptions in the question,
- prefer concise and direct answers,
- abstain ("I do not know") when uncertainty remains.

Compared to mild rewriting, this prompt allows more aggressive modification and supports abstention to avoid hallucinated outputs.

1.3.4 Dataset-Specific Adaptation

TruthfulQA

For TruthfulQA, mitigation explicitly targets common misconceptions and aims to correct semantically incorrect answers. Unlike NQ, answers often require rewriting rather than extraction, making semantic correction essential.

In the implementation, mitigation is realized through a candidate-based correction process. Multiple candidate answers are generated using a combination of mild rewriting, strict correction, and edit-based refinement prompts.

These candidates are subsequently filtered to remove low-quality outputs (e.g., overly long or uninformative responses), and grouped into answer and abstention types. The final selection is performed using detector-based scoring and self-consistency signals, as described in the main methodology.

Additionally, an abstention mechanism allows the system to output "I do not know" when all candidates are considered unreliable.

FEVER

For FEVER, mitigation is tightly coupled with evidence-grounded verification. Prompts explicitly constrain the model to rely only on the provided evidence and to produce structured outputs with a label and a short justification.

In addition, optional span-based evidence extraction may be applied to further encourage grounding in specific textual fragments.

Further consistency is enforced through model-based filtering and reranking, as described in the main methodology, including contradiction detection via NLI and label-consistent grouping.

Natural Questions (NQ)

For NQ, mitigation follows an extractive-dominant strategy that emphasizes short, context-grounded answers. Candidate answers are primarily derived from short spans in the local context using question–context overlap heuristics.

These extractive candidates may be supplemented by a small number of short rewritten answers under relaxed constraints. All candidates are filtered based on length and context alignment, and the final selection is performed using detector-based scoring and self-consistency signals, as described in the main methodology. In the current implementation, abstention is not enabled for NQ.

Appendix B

Example Outputs

Appendix C

Additional Results